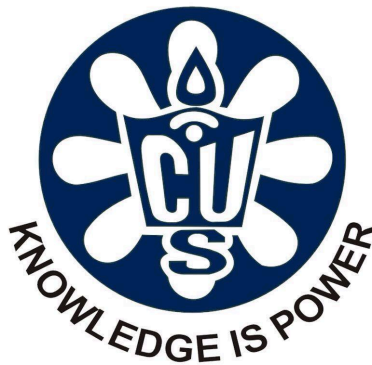


A
PROJECT REPORT
ON

“ TechVolt - Electronic Components
E-Commerce Website ”

DEVELOPED AT:



**SHREE C. U. SHAH COLLEGE OF MANAGEMENT &
COMPUTER EDUCATION, WADHWAN**

SUBMITTED BY:

Bhavik Kher (24CS002UG02412)
Ankit Chavda (24CS002UG02381)

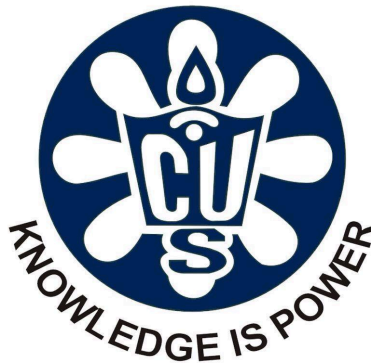
Affiliated to:
Saurashtra University

“TechVolt - Electronic Components E-Commerce Website”

Project Logo



SUBMITTED TO:



**SHREE C. U. SHAH COLLEGE OF MANAGEMENT &
COMPUTER EDUCATION, WADHWAN**

Affiliated to:
Saurashtra University

GUIDED BY:

Dr.Viral Dagli

DEVELOPED BY:

Bhavik Kher(24CS002UG02412)
Ankit Chavda(24CS002UG02381)

**ACADEMIC YEAR: 2026-27
BCA SEM – 5**

CANDIDATE’S DECLARATION

I, **Bhavik Kher.** declare that final year report entitled “**TechVolt - Electronic Components E-Commerce Website**” is my own work conducted under the supervision of **Dr.Viral Dagli,** the project guide of “TechVolt - Electronic Components E-Commerce Website”.

I further declare that to the best of my knowledge the report for BCA 5th semester does not contain part of the work which has been submitted for the eligible for BCA degree either in this or any other university without proper citation.

Submitted To:

Dr. Viral Dagli

Department of Computer Science,

C. U. Shah College. Wadhwan City,

State: Gujarat.

Bhavik Kher

Branch: Computer Application

CANDIDATE'S DECLARATION

I, **Ankit Chavda**, declare that final year report entitled “**TechVolt - Electronic Components E-Commerce Website**” is my own work conducted under the supervision of **Dr.Viral Dagli**, the project guide of “TechVolt - Electronic Components E-Commerce Website”.

I further declare that to the best of my knowledge the report for BCA 5th semester does not contain part of the work which has been submitted for the eligible for BCA degree either in this or any other university without proper citation.

Submitted To:

Dr. Viral Dagli

Department of Computer Science,

C. U. Shah College. Wadhwan City,

State: Gujarat.

Ankit Chavda

Branch: Computer Application



JAIN CITIZENS' EDUCATION MANAGED
SHREE C. U. SHAH COLLEGE OF COMMERCE, MANAGEMENT &
COMPUTER EDUCATION, WADHWAN

Affiliated to: Saurashtra University no. 2/957/96

Dt.: 07/08/1996

College Code: 29003

This is to certify that Mr. **Bhavik Kher** of TYBCA SEM 5 has satisfactorily completed his / her academic project work report on **TechVolt - Electronic Components E-Commerce Website** as a part of course prescribed by Saurashtra University for the academic year 2026 – 27.

Enrollment No.: **24CS002UG02412**

Date: 28 September 2026

Project Guide

(Dr. Viral R. Dagli)



JAIN CITIZENS' EDUCATION MANAGED
SHREE C. U. SHAH COLLEGE OF COMMERCE, MANAGEMENT &
COMPUTER EDUCATION, WADHWAN

Affiliated to: Saurashtra University no. 2/957/96

Dt.: 07/08/1996

College Code: 29003

This is to certify that Mr. **Ankit Chavda** of TYBCA SEM 5 has satisfactorily completed his / her academic project work report on **TechVolt - Electronic Components E-Commerce Website** as a part of course prescribed by Saurashtra University for the academic year 2026 – 27.

Enrollment No.: **24CS002UG02381**

Date: 28 September 2026

Project Guide

(Dr. Viral R. Dagli)

ACKNOWLEDGEMENT

As of now of time when we are composing this documentation we had an opportunity to look four months back when we were at this point to take up any project close by taking into consideration this cut of time and the status now, we unequivocally feel a feeling of distinction, a feeling of having developed. We consider our self-lucky to be served this excellent piece of information and experience for four valuable months we spent at Shree C. U. Shah College of Management & Computer Education, Wadhwan.

For this, we are very thankful to our college that provided their resources in making us competent and deserving of taking up such a difficult undertaking. For this undertaking project that needs an additional knowledge and guidance for the successful development, we say thanks to Dr.Viral Dagli, Department of B.C.A division.

ABSTRACT

TechVolt – Electronic Components E-Commerce Website is a web-based application designed to provide a convenient platform for purchasing electronic components online. The system is developed for students, hobbyists, engineers, and electronics enthusiasts who need easy access to quality electronic products. The website offers a wide range of components, including Arduino Uno, Arduino Nano, ESP32, NodeMCU, Raspberry Pi, sensors, resistors, capacitors, ICs, motors, displays, power modules, and other electronic accessories. The application provides a simple and user-friendly interface where customers can register, log in, browse products by category, search for specific items, view product details, add products to the shopping cart, and place orders. Customers can also manage their profiles and view their order history. The website includes a secure admin panel that enables administrators to manage products, categories, inventory, customer information, and orders efficiently. The admin can add, edit, or remove products, update stock quantities, and monitor sales through the system. The project is developed using HTML, CSS, JavaScript, PHP, and MySQL, with Wampp used as the local server environment. The database stores all information related to users, products, categories, orders, and inventory. The primary objective of TechVolt is to simplify the buying and selling of electronic components through an online platform while improving inventory management and customer experience. This project demonstrates web development, database management, user authentication, CRUD operations, and e-commerce functionality. Future enhancements may include online payment integration, order tracking, product reviews, and AI-based product recommendations.

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INTRODUCTION

1.1 PROJECT DETAILS

Project Title: TechVolt – Electronic Components E-Commerce Website

TechVolt is a web-based e-commerce platform developed for buying and managing electronic components online. The main purpose of this project is to provide a convenient platform where students, hobbyists, engineers, electronics enthusiasts and other customers can browse and purchase different types of electronic components.

The system allows users to register and log in to their accounts, browse available products, search for required electronic components, view detailed product information, add products to the cart or wishlist, and place orders. Users can also manage their profile and view their previous orders and order details.

The website provides different categories of electronic components such as Accessories, Displays, ICs, Microcontrollers, Motors & Actuators, Sensors, Power Modules, Passive Components and Single Board Computers. Each product contains important information such as product name, brand, model number, price, stock availability, description, images and specifications.

TechVolt also includes an administrative panel for managing the complete e-commerce system. The administrator can manage products, categories, inventory, customer information, orders and return requests. The admin panel also provides a dashboard and reports to help the administrator monitor the overall activities of the website.

The project is developed using **HTML, CSS, JavaScript, PHP and MySQL**. The system is implemented on a local WAMP server and uses a database to store and manage customer, product, order, inventory and return-related information.

1.2 PURPOSE

The main purpose of the TechVolt project is to develop a simple, organized and user-friendly online platform for purchasing electronic components.

In traditional methods, customers may need to visit different shops or search through multiple sources to find the required electronic components. It can be difficult to compare products, check availability and maintain information about previous purchases. TechVolt aims to provide these facilities through a single web-based platform.

The project provides customers with an easy way to search and explore electronic components, check product details and stock availability, add products to their cart or wishlist and place orders online.

The administrative section of the system helps the administrator manage products and categories, maintain stock information, monitor customer orders, handle return requests and view sales-related reports. This makes the overall management of the e-commerce website more organized and efficient.

The project is mainly developed to demonstrate the practical implementation of web development concepts such as frontend design, backend programming, database management, user authentication, product management, shopping cart functionality and order management.

1.3 SCOPE

The scope of TechVolt is to provide an online platform for purchasing electronic components.

The system allows customers to browse products from different categories. Users can search for products and view their details, specifications, price and availability. Customers can add products to their cart and wishlist. The system allows users to place orders and view their order history.

The admin panel provides complete management of products and categories. The administrator can manage inventory and update product stock. The admin can view and manage customer information and orders. The system also provides return request management for customers.

Sales and order reports are available for administrative purposes. The project can be further extended with online payment integration, product reviews and notifications.

1.4 OBJECTIVES

The main objectives of the TechVolt project are:

- To develop an online electronic components shopping website.
- To provide a simple and user-friendly interface.
- To allow users to register and securely log in.
- To provide product categories and search functionality.
- To display product details and specifications.
- To provide cart and wishlist functionality.
- To allow users to place orders online.
- To maintain user order history.
- To manage products and categories through the admin panel.
- To manage inventory and stock availability.
- To manage customer orders and return requests.
- To provide sales and order reports.

PROJECT MANAGEMENT

2.1 FEASIBILITY STUDY:

2.1.1 Technical Feasibility:

Technical feasibility checks whether the required technology and resources are available to develop and run the **TechVolt – Electronic Components E-Commerce Website**. The project can be developed using commonly available technologies such as **PHP, MySQL, HTML, CSS, and JavaScript**. The system can run on a normal computer using **WAMP Server**, which provides the web server and database environment. The required hardware, software, and development tools are easily available and do not require high-cost equipment. Therefore, the project is technically feasible and can be developed, maintained, and used successfully with the available technology.

2.1.2 Operational Feasibility:

Operational feasibility checks whether the TechVolt system is easy to use and suitable for the daily activities of customers and administrators. The system provides simple and user-friendly features for browsing electronic components, searching products, managing cart and wishlist, placing orders, and viewing order information. The admin can easily add, update, delete, and view products, categories, inventory, customers, orders, and return requests. The system reduces manual work and makes electronic component shopping and management easier and faster. Therefore, the system is operationally feasible and can be easily used by both customers and administrators.

2.1.3 Economical Feasibility:

Economical feasibility checks whether the TechVolt system can be developed within a reasonable cost. The project uses technologies such as PHP, MySQL, HTML, CSS, JavaScript, and WAMP Server, which are easily available and require very little or no additional software cost. The system can be developed and operated on a normal computer without expensive hardware. It also reduces manual work and provides organized management of products, orders, inventory and customer information. Therefore, the project is economically feasible and can be developed at a low cost.

2.2 PROJECT PLANNING:

2.2.1 Project Development Approach & Justification:



How use this model for developing this project

- **In the initiation phase**, the main goals, scope, requirements, and resources of the TechVolt project are identified. The requirements of customers and administrators are studied to decide the major features of the system, such as product browsing, product search, cart, wishlist, order management, inventory management, and return management. This phase provides a clear direction for the project.

- **In the planning phase**, the steps required to achieve the project goals are decided. The required technologies, database tables, website pages, admin modules, development tasks, and project schedule are planned. The resources required for development and testing are also identified.
- **Executing a project** means putting the planned development activities into action and completing the different modules of TechVolt. Generally, this includes developing the customer side, admin panel, database and required functionalities. Specific steps include:
 - Developing customer-side web pages.
 - Developing the admin panel and management modules.
 - Creating and connecting the MySQL database.
 - Implementing product, cart, wishlist and order functionality.
 - Implementing inventory and return management.
 - Testing different modules during development.
 - Making required changes and correcting errors.
- **In the closing phase of the project management lifecycle**, you will conclude all development activities and prepare the completed TechVolt system for final submission. The completed website and admin panel will be reviewed to ensure that the required features are working properly. The final documentation, screenshots, diagrams and testing results will also be completed. Steps in the closing phase can include:
 - Reviewing the completed TechVolt system and its functionalities.
 - Testing the final system and correcting remaining errors.
 - Reviewing the project requirements and completed modules.
 - Preparing the final project report and documentation.
 - Taking screenshots of the completed customer and admin pages.
 - Preparing the final project for submission and presentation.
 - Reviewing the overall project work and noting possible improvements for future versions.

2.2 PROJECT SCHEDULING:

Scheduling in project management is the process of listing the activities, tasks and milestones required to complete a project. A project schedule normally includes the activities to be performed, their expected duration and the resources required for development.

For the TechVolt project, scheduling helps to divide the development work into different stages such as requirement analysis, database design, frontend development, backend development, admin panel development, testing and documentation. Proper scheduling helps to complete the project activities in an organized manner and within the available time.

Resources required for Development of Project:

- Human effort
- Computer or laptop
- Sufficient disk space
- WAMP Server
- PHP and MySQL
- HTML, CSS and JavaScript
- VS Code
- Internet connection
- Time required for development and testing

Advantages of Project Scheduling:

- It ensures that project tasks are completed in a planned manner.
- It helps in managing development activities and deadlines.
- It helps to identify pending tasks and possible delays.
- It helps in proper utilization of available resources.
- It helps to monitor the progress of the project.
- It helps in planning testing and documentation activities.

SYSTEM REQUIREMENT STUDY

3.1 STUDY OF CURRENT SYSTEM

In the current system, customers generally need to visit physical electronic component stores or use different online sources to find the required products. Product information, price and stock availability may not always be available at one place. Managing products, customer information, orders and inventory manually can also require additional effort. The TechVolt system is developed to provide a centralized web-based platform for electronic component shopping and management. Customers can browse different products, search for required components, view product details, add products to cart or wishlist and place orders. The administrator can manage products, categories, inventory, customers, orders and return requests through the admin panel.

3.2 PROBLEMS AND WEAKNESS OF CURRENT SYSTEM

The existing/manual system may have several problems while managing electronic component purchasing and sales activities. Some common problems are:

- Customers may need to visit different stores to find required components.
- Product information may not be available in one centralized place.
- Checking product availability can take additional time.
- Manual order management can increase the possibility of errors.
- Maintaining product and inventory information manually is difficult.
- Customer and order records may be difficult to organize.
- Searching for a particular electronic component can be time-consuming.
- Managing return requests manually can be difficult.
- Sales and order information may not be available in an organized report.

TechVolt addresses these problems by providing an online platform where customers and administrators can manage their respective activities in an organized manner.

3.3 HARDWARE AND SOFTWARE REQUIREMENTS:

Hardware Requirements (Minimum):

Component	Specification
Processor	Intel Core 2 Duo
RAM	4 GB RAM
Hard Disk	500 GB

Software Requirements

Operating System: Windows 7, Windows 8.1, etc...

Front End Tools: HTML, CSS and JavaScript

Back End Tools: PHP and MySQL

Development Server: WAMP Server

Development Tool: Visual Studio Code

Web Browser: Google Chrome / Microsoft Edge

Office Automation Tools:

- Microsoft Office Word
- Microsoft Visio

REQUIREMENT OF PROPOSED SYSTEM

4.1 MAIN MODULES OF THE SYSTEM:

There are basically two main modules in the TechVolt system:

- Admin User
- Registered User

The Admin User manages the overall website and its activities, while the Registered User can access the shopping features of the TechVolt website after successful registration and login.

4.2 MODULE DESCRIPTION:

There are two main modules in this system.

1) Admin User:

The Admin User manages the overall activities of the TechVolt website. The administrator can manage products, categories and inventory. The admin can add, edit and delete products and update product stock. The admin can also manage customers, orders and return requests. The dashboard provides information about sales, orders, products and low-stock items. The admin can also view reports related to sales and order activities.

2) Registered User:

The Registered User can access the shopping features of TechVolt after successful registration and login. The user can browse and search electronic components, view product details and specifications, and check product availability. The user can add products to the cart and wishlist and place orders through the checkout system. The user can also manage profile information and view previous orders and order details.

4.3 FEATURES OF NEW SYSTEM:

The main features of the proposed **TechVolt** system are:

- User registration and secure login.
- Admin login and management panel.
- Product browsing and category-wise product display.
- Product search functionality.
- Product details and specifications.
- Product price and stock availability.
- Shopping cart management.
- Wishlist management.
- Checkout and order placement.
- Cash on Delivery and online payment options.
- Order confirmation and order history.
- Order details and order status tracking.
- Customer profile management.
- Product and category management by admin.
- Inventory and stock management.
- Customer management.
- Return request management.
- Sales and order reports.
- Product Active/Inactive status management.
- MySQL database management.

5.1 SYSTEM ARCHITECTURE DESIGN:

Figure 5.1: $E - R$ Diagram

5.1.2 Use case Diagram:

a) Admin

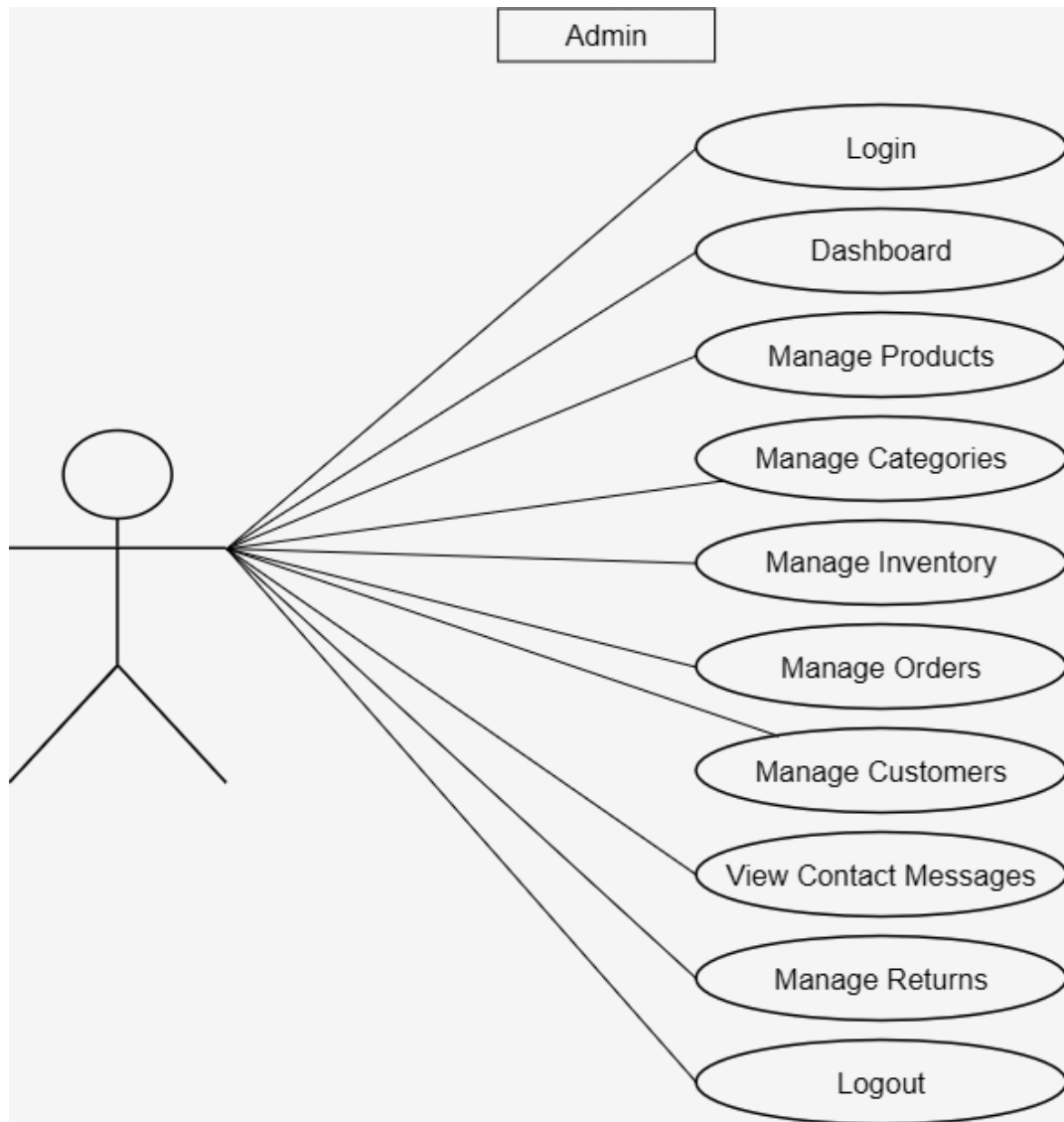


Figure 5.2: Use case Diagram for Admin

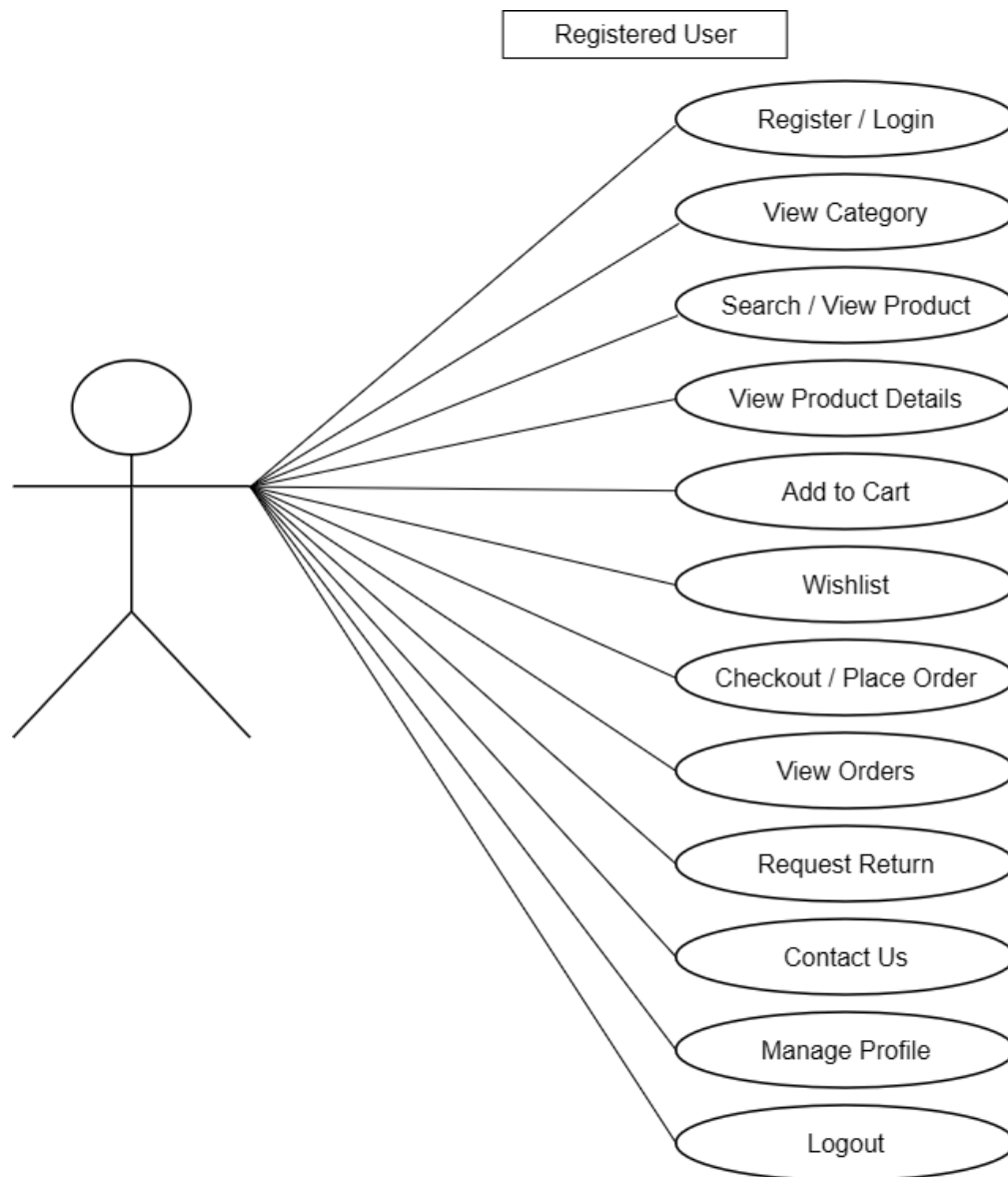
b) Register:

Figure 5.3: Use case Diagram for Register

5.1.3 Data Flow Diagram:

a) Context Diagram :

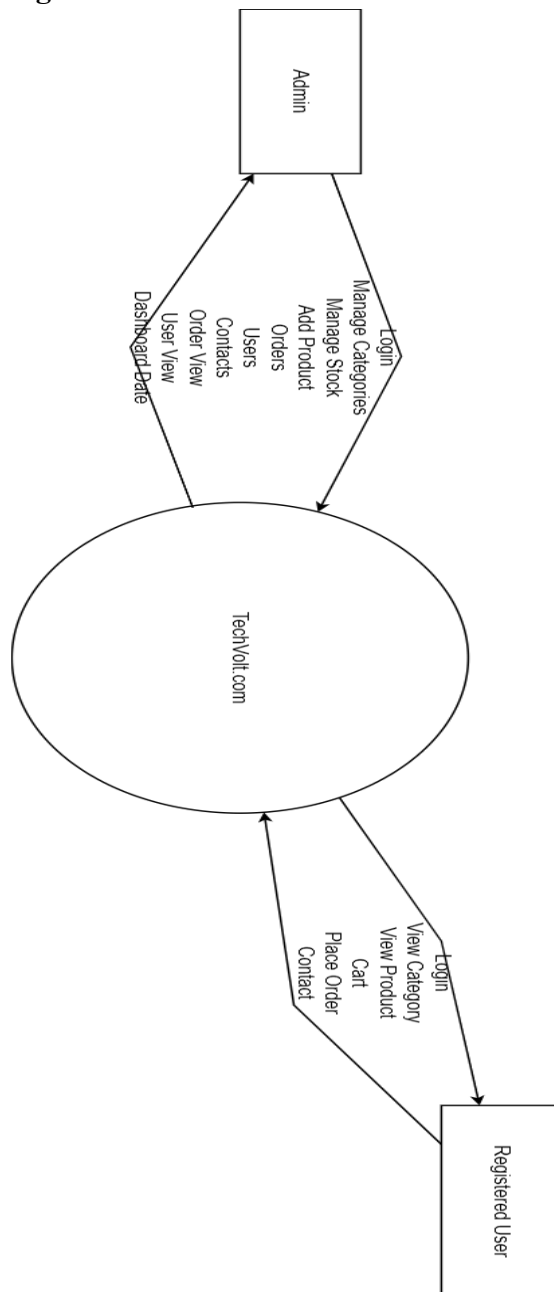


Figure 5.6: Context Diagram

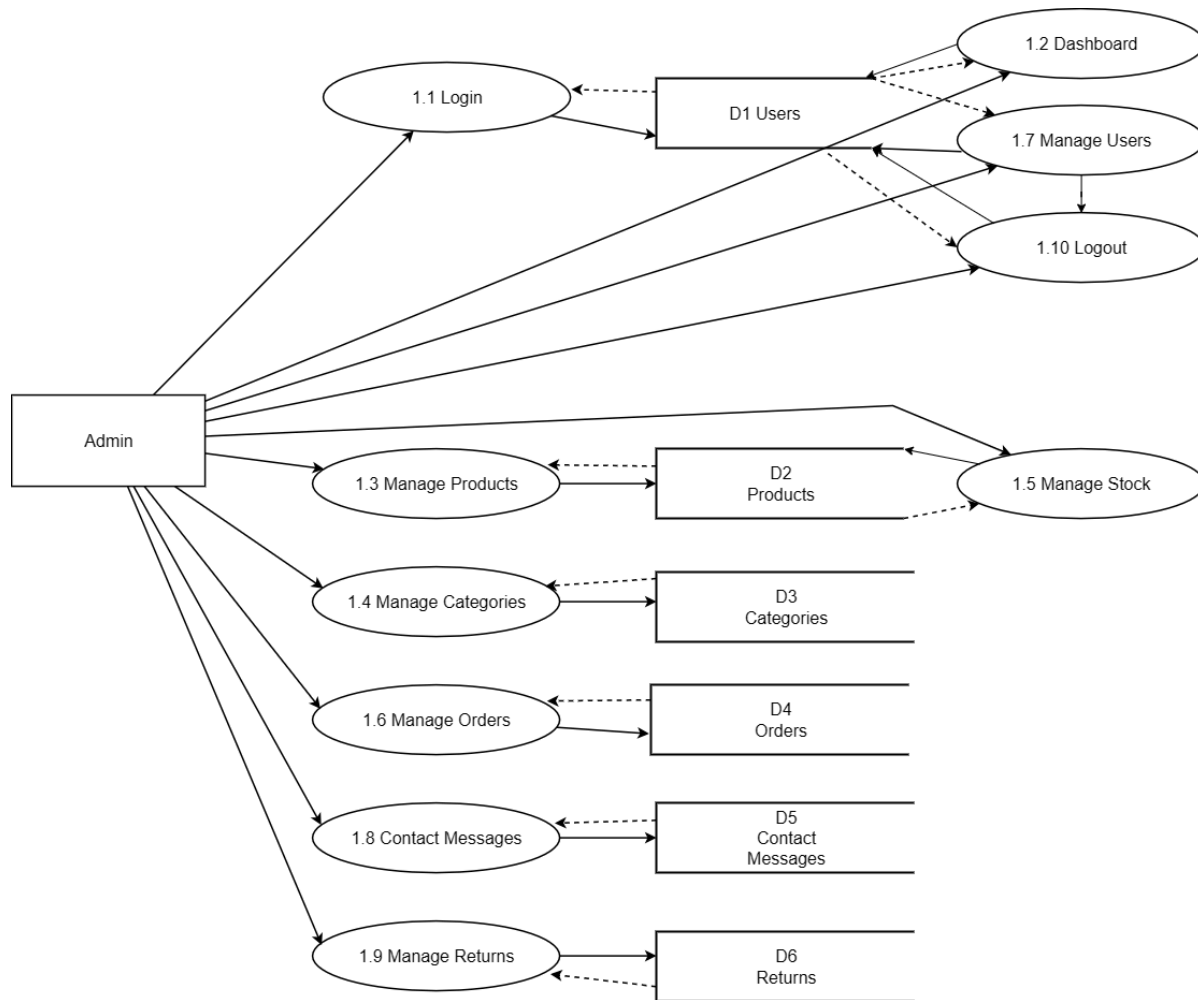
Level 1 Data Flow Diagram :**a) Admin**

Figure 5.6: 1 - Level Data Flow Diagram (admin)

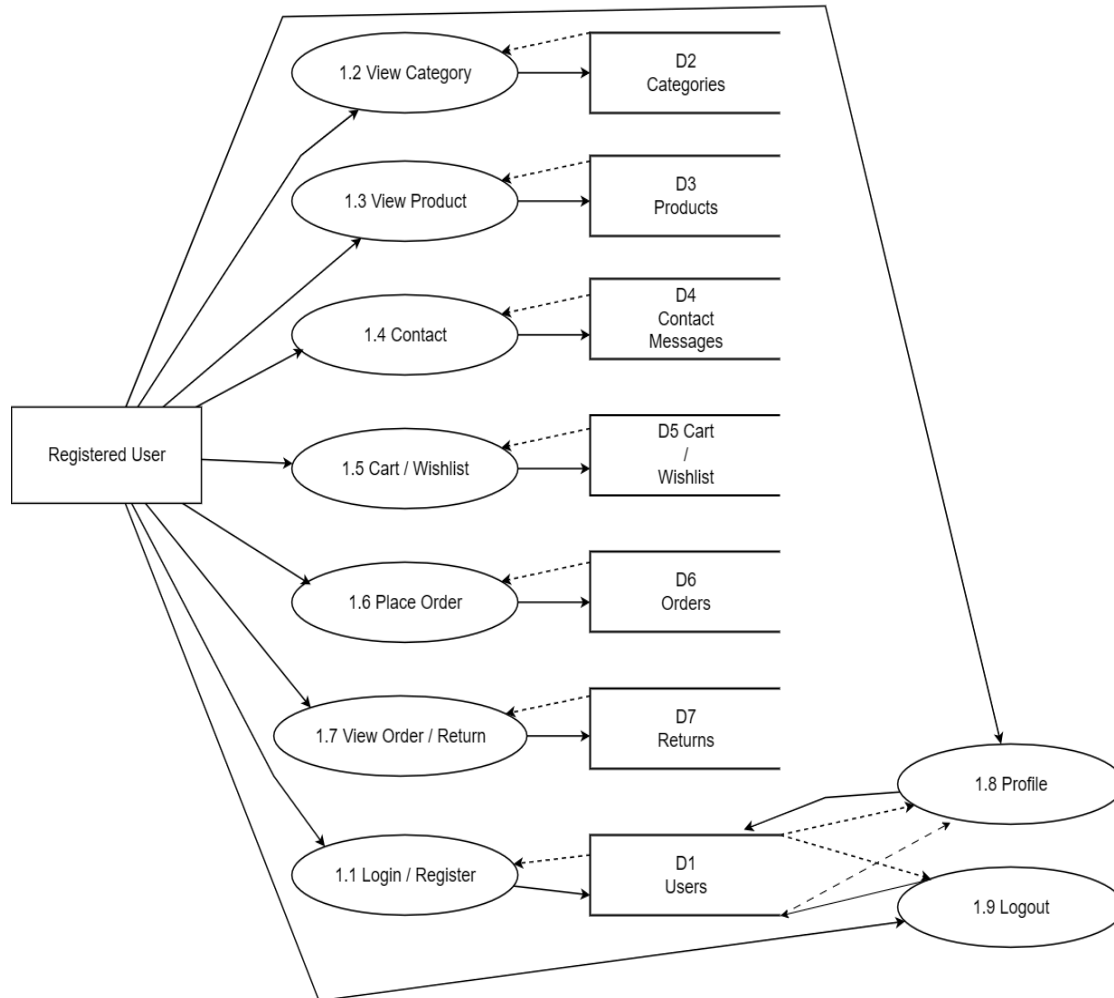
a) Registered User

Figure 5.6: 1 - Level Data Flow Diagram (user)

5.1.4 Activity Diagram :

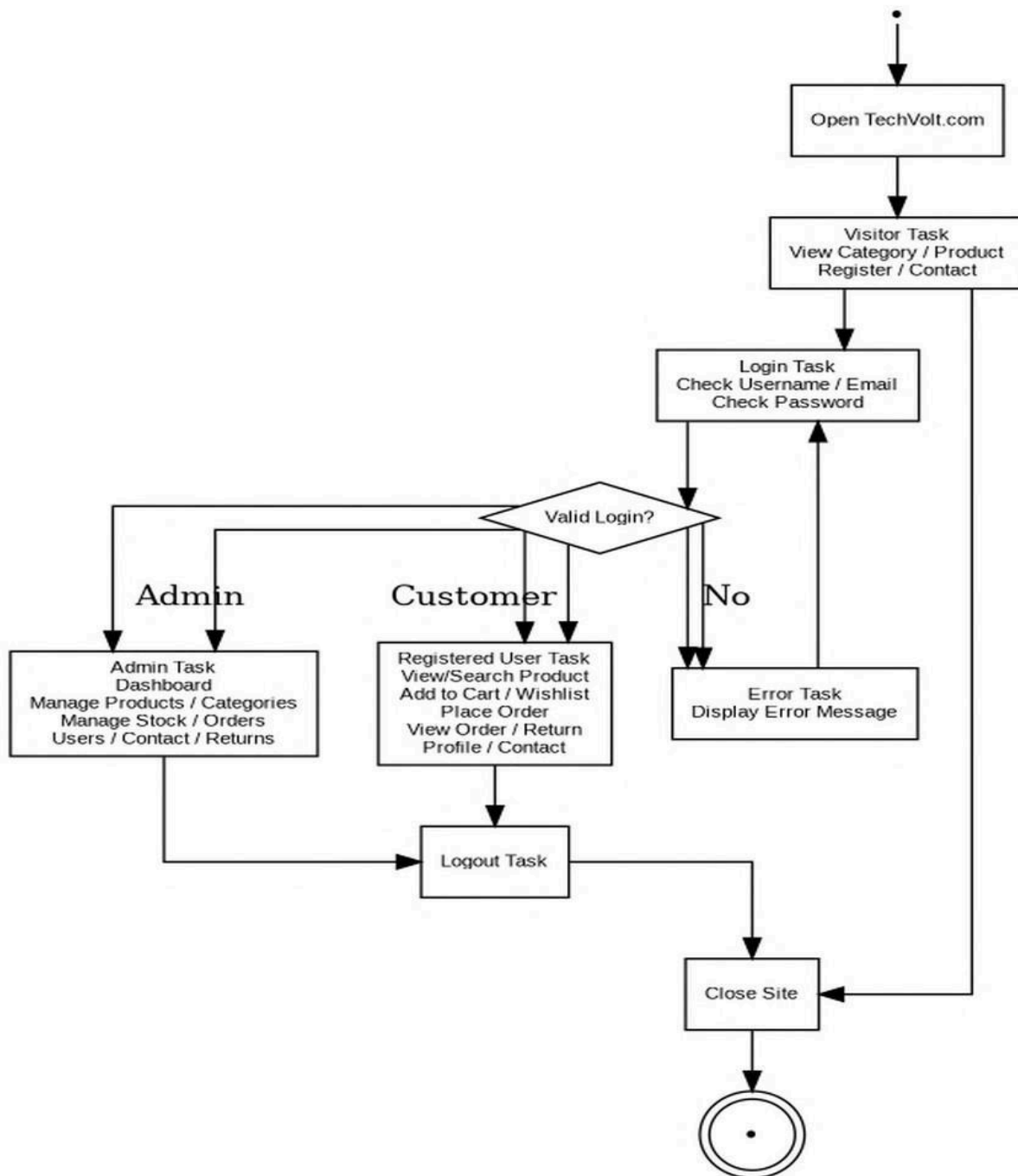


Figure 5.7: Activity Diagram

• 5.2 DATA DICTIONARY

Table Name: users

Description: This table contains the Users details and Admin detail.

SR. No	FIELD NAME	DATA TYPE	SIZE	CONSTRAINT	DESCRIPTION
1	user_id	Int	5	Primary key	Id of admin
2	full_name	Varchar	100	Not null	Full name of Users
3	username	Varchar	50	Not null,	Password of users
4	email	Varchar	150	Not Null	Email of users
5	password	Varchar	255	Not Null	Password of users
6	profile_image	Varchar	255	Null	profile_image path of users
7	role	enum	-	-	role of users like admin or customer
8	status	enum	-	-	Status of user like active or inactive
9	created_at	timestamp	-	default current timestamp	date and time of created user
10	updated_at	timestamp	-	on update current timestamp	date and time of updated user information

Table Name: products

Description: This table contains Products details.

SR. No	FIELD NAME	DATA TYPE	SIZE	CONSTRAINT	DESCRIPTION
1	product_id	Int	5	Primary key	Id of products
2	product_name	Varchar	150	Not null	Product name for each products
3	category_id	Int	5	Not null	category_id for products
4	brand	Varchar	100	Null	brand name for products
5	model_number	Varchar	100	Null	Model number for products
6	sku	Varchar	100	Not Null	Sku for products
7	short_description	Varchar	500	Null	Short_description for products
8	description	text	-	Not Null	Description for products
9	price	Int	5	Not Null	Price for products
10	stock	Int	5	Not Null	Stock for products
11	rating	decimal	2	Not Null	Rating for products
12	review_count	Int	5	Not Null	Review_count for products
13	status	enum	-	Not Null	Status for products default active
14	featured	tinyint	-	Not Null	Product is featured or not
15	created_at	timestamp	-	Not Null	Current time and date for products

Table Name: product_specifications

Description: This table contains Product Specification details.

SR. No	FIELD NAME	DATA TYPE	SIZE	CONSTRAINT	DESCRIPTION
1	specification_id	Int	5	Primary key	Specification Id of products
2	product_id	Int	5	Not null	Id of products
3	spec_name	Varchar	100	Not null	Specification name for product
4	spec_value	Varchar	255	Not null	Specification value for product
5	sort_order	Int	5	Not null	Sort order for specification

Table Name: product_images

Description: This table contains Product Images details.

SR. No	FIELD NAME	DATA TYPE	SIZE	CONSTRAINT	DESCRIPTION
1	image_id	Int	5	Primary key	Image id for products
2	product_id	Int	5	Not null	Id of products
3	image_path	Varchar	255	Not null	Image Path for products
4	alt_text	Varchar	200	Null	Alter text for product image
5	is_primary	tinyint	1	Not null	Set a primary image to specific product
6	sort_order	Int	5	Not null	Sort order for images
7	created_at	timestamp	-	Not null	Current time and date for product image

Table Name: Orders

Description: This table contains Order details.

SR. No	FIELD NAME	DATA TYPE	SIZE	CONSTRAINT	DESCRIPTION
1	order_id	Int	5	Primary key	Order id for Orders
2	user_id	Int	5	Not null	User id for orders
3	address_id	Int	5	Not null	Address id for orders
4	subtotal	decimal	10,2	Not null	Subtotal for order
5	delivery_charge	decimal	10,2	Not null	Delivery charge for orders
6	gst_amount	decimal	10,2	Not null	Gst amount for orders
7	total_amount	decimal	10,2	Not null	Total amount for orders
8	payment_method	enum	-	Not null	Payment method for orders
9	payment_status	enum	-	Not null	Payment status for orders
10	order_status	enum	-	Not null	Order status for orders
11	shipping_name	Varchar	100	Not null	Shipping name for orders
12	shipping_phone	Varchar	15	Not null	Shipping phone for orders
13	shipping_address	Varchar	255	Not null	Shipping address for order
14	created_at	timestamp	-	Not null	date and time of orders

Table Name: order_items

Description: This table contains Order Items details.

SR. No	FIELD NAME	DATA TYPE	SIZE	CONSTRAINT	DESCRIPTION
1	order_item_id	Int	5	Primary key	Order_items_id for orders
2	order_id	Int	5	Not null	Order id for order item
3	product_id	Int	100	Not null	Product id for order item
4	product_name	Varchar	150	Not null	Product name for order item
5	price	Int	5	Not null	Price for order items
6	quantity	Int	5	Not null	Quantity for order items
7	subtotal	Int	5	Not null	Subtotal for order items

Table Name: categories

Description: This table contains categories details.

SR. No	FIELD NAME	DATA TYPE	SIZE	CONSTRAINT	DESCRIPTION
1	category_id	Int	5	Primary key	Category_id for all categories
2	category_name	Varchar	100	Not null	Category_name for categories
3	slug	Varchar	120	Not null	Slug for category
4	description	text	255	Null	Description for category
5	status	enum	-	Not null	Status for category
6	created_at	timestamp	-	Not null	Current time and date for category

Table Name: carts

Description: This table contains carts details.

SR. No	FIELD NAME	DATA TYPE	SIZE	CONSTRAINT	DESCRIPTION
1	cart_id	Int	5	Primary key	Cart Id of specific user
2	user_id	Int	5	Not null	User Id of cart
3	created_at	timestamp	-	Not null	Current time and date for cart
4	updated_at	timestamp	-	Null	On update Current time and date for cart

Table Name: cart_items

Description: This table contains Cart Items details.

SR. No	FIELD NAME	DATA TYPE	SIZE	CONSTRAINT	DESCRIPTION
1	cart_item_id	Int	5	Primary key	Cart_item_id for cart_items
2	cart_id	Int	5	Not null	Cart_id for cart_item
3	product_id	Int	5	Not null	Product_id for cart_item
4	quantity	Int	5	Not null	Quantity for cart_item
5	created_at	timestamp	-	Not null	Current time and date for cart_item

Table Name: addresses

Description: This table addresses of users details.

SR. No	FIELD NAME	DATA TYPE	SIZE	CONSTRAINT	DESCRIPTION
1	address_id	Int	5	Primary key	Address_id for addresses
2	user_id	Int	5	Not null	User-id for addresses
3	full_name	Varchar	100	Not null	Full name for address
4	phone	Varchar	15	Not null	Phone number for address
5	address_line	Varchar	255	Not null	Address_line for address
6	city	Varchar	100	Not null	City name for address
7	state	Varchar	100	Not null	State name for address
8	pincode	Varchar	10	Not null	Pincode for address

Table Name: wishlist

Description: This table contains wishlist product details.

SR. No	FIELD NAME	DATA TYPE	SIZE	CONSTRAINT	DESCRIPTION
1	wishlist_id	Int	5	Primary key	Wishlist Id of specific user
2	user_id	Int	5	Not null	User Id of wishlist
3	product_id	Int	5	Not null	Product_id for wishlist
4	created_at	timestamp	-	Not null	Current time and date for wishlist
5	updated_at	timestamp	-	Not null	On update Current time and date for wishlist

Table Name: contact_messages

Description: This table contains contact Messages details.

SR. No	FIELD NAME	DATA TYPE	SIZE	CONSTRAINT	DESCRIPTION
1	message_id	Int	5	Primary key	message_Id of messages
2	user_id	Int	5	Null	User Id of messages
3	name	Varchar	-	Not null	Name for messages
4	email	Varchar	-	Null	Email for messages
5	subject	Varchar			Subject for messages
6	message	text	-		Main message for contact admin
7	created_at	timestamp			Current time and date for Messages

Table Name: returns

Description: This table contains returns details.

SR. No	FIELD NAME	DATA TYPE	SIZE	CONSTRAINT	DESCRIPTION
1	return_id	Int	5	Primary key	return_Id of returns
2	order_item_id	Int	5	Not null	Order_item_id for returns
3	user_id	Int	5	Not null	User Id for returns
4	reason	Varchar	255	Not null	Reason for returns
5	description	text	-	Null	Description for returns
6	return_status	enum	-	Not null	Return status for return
7	created_at	timestamp	-	Not null	Current time and date for returns
8	updated_at	timestamp	-	Null	On update Current time and date for returns

5.3 Samplecode

Login Design:



5.3.1 Login page

Code: Login.php

```
<?php
session_start();
require_once "includes/database.php";
$alert = "";

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $login = trim($_POST["login"]);
    $password = $_POST["password"]; if (
        empty($login) || empty($password)
    ) {
        $alert = "
```

```
Swal.fire({
    icon: 'warning',
    title: 'Missing Information',
    text: 'Please fill in all required fields.', timer:
    2000,
    theme: 'dark'
});
";
} else {
    $sql = "SELECT * FROM users WHERE username = ? OR email = ?";
    $stmt = mysqli_prepare($conn, $sql); if
    (!$stmt) {
        die("Prepare failed: " . mysqli_error($conn));
    }
    mysqli_stmt_bind_param($stmt, "ss", $login, $login); mysqli_stmt_execute($stmt);
    $result = mysqli_stmt_get_result($stmt); if
    (mysqli_num_rows($result) > 0) {
        $user = mysqli_fetch_assoc($result);
        if (password_verify($password, $user["password"])) { if
            ($user["status"] == "active") {
                session_regenerate_id(true);
                $_SESSION["user_id"] = $user["user_id"];
                $_SESSION["username"] = $user["username"];
                $alert = "
                    Swal.fire({
```

```
        icon: 'success',

        title: 'Login Successful.',

        text: 'You have logged in to your TechVolt account successfully.', timer: 2000,

        showConfirmButton: false,

        theme: 'dark'

    }).then(() => {

        window.location.href = 'index.php';

    });

    ";

} else {

    $alert = "

    Swal.fire({

        icon: 'error',

        title: 'Login Unsuccessful.',

        text: 'Your account is Suspended.',

        timer: 2000,

        showConfirmButton: false,

        theme: 'dark'

    }).then(() => {

        window.location.href = 'index.php';

    });

    ";

}

} else {

    $alert = "
```

```
        Swal.fire({
            icon: 'error',
            title: 'Incorrect Password',
            text: 'Please make sure your password is correct.', timer: 2000,
            theme: 'dark'
        });
    ";
}
} else {
    $alert = " Swal.fire({
        icon: 'error',
        title: 'Login Failed',
        text: 'We could not find an account with that username or email.', theme: 'dark'
    });
    ";
}
}
}
?>
<!DOCTYPE html>

<html lang="en">
<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Login to TechVolt</title>

    <link rel="stylesheet" href="assets/css/login.css">
```

```
<script src="https://cdn.jsdelivr.net/npm/sweetalert2@11"></script>

</head>
<body>
<div class="form-container">
  <h2>Login</h2>

  <form action="login.php" method="post">

    <div class="input-box">
      <input id="username" name="login" type="text" placeholder="">

      <label for="username" class="username-or-password">Username OR Email</label>
    </div>

    <div class="input-box">
      <input id="password" name="password" type="password" placeholder="">

      <label for="password">Password</label>
    </div>

    <button type="submit" class="submit-btn">Login</button>

    <div class="extra-link">
      <p>Don't have an account?</p>

      <a class="register" href="register.php">Create Account &#10132;</a>

      <br>

      <br>

      <a class="admin" href="Admin/admin-login.php">Admin Login &#10132;</a>
    </div>
  </div>

</form>
<section>

  
```

<h3>Power Your Ideas. Build Your Future.</h3>

<p>Your one-stop destination for electronic components, development boards, sensors, modules, and more.</p>

</section>

<!-- PHP Alert -->

<?php if (!empty(\$alert)): ?>

<script>

 window.addEventListener('DOMContentLoaded', function() {

 <?php echo \$alert; ?>

 });

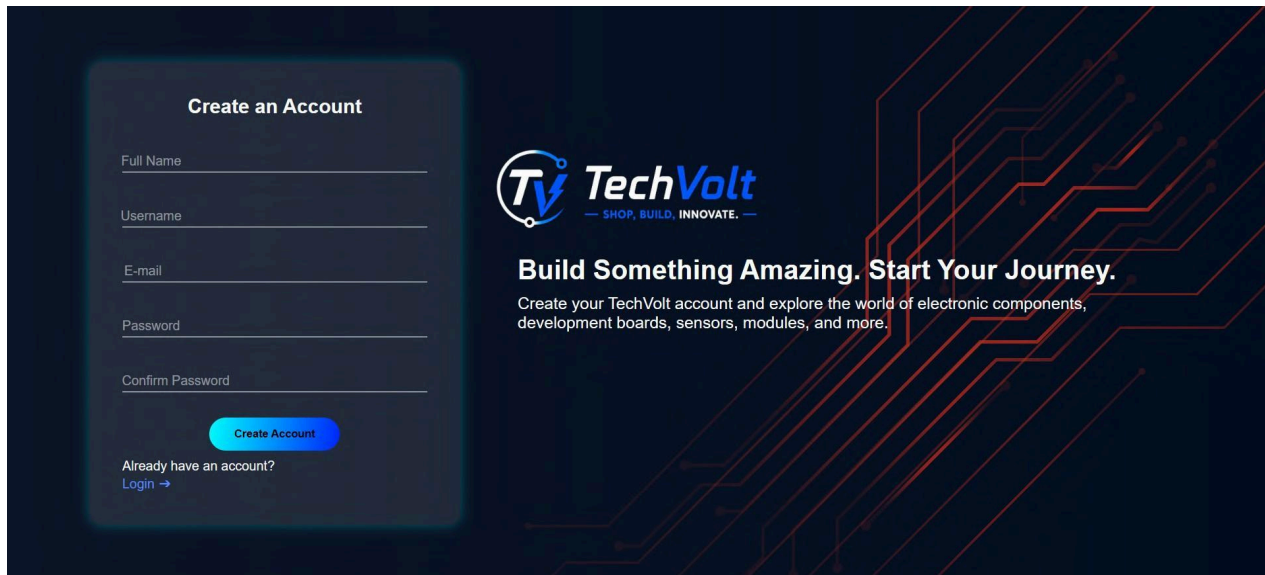
</script>

<?php endif; ?>

</body>

</html>

Registration Design:

The image shows a registration page design for a company called TechVolt. The page has a dark blue background with a subtle pattern of red circuit lines. On the left, there is a white rounded rectangle containing a registration form. The form has the title "Create an Account" and five input fields: "Full Name", "Username", "E-mail", "Password", and "Confirm Password". Below these fields is a blue "Create Account" button. At the bottom of the form, there is a link that says "Already have an account? Login →". To the right of the form, the TechVolt logo is displayed, which consists of a stylized "TV" in a circle followed by the text "TechVolt" and the tagline "— SHOP, BUILD, INNOVATE. —". Below the logo, the text "Build Something Amazing. Start Your Journey." is written in a bold font. Underneath this, a smaller line of text says "Create your TechVolt account and explore the world of electronic components, development boards, sensors, modules, and more."

5.4.1 Registration Page

Code: Registration.php

`<?php`

```
require_once "includes/database.php";
$alert = "";
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $fullName = trim($_POST["fullName"]);

    $username = trim($_POST["username"]);

    $email = trim($_POST["email"]);

    $password = $_POST["password"];

    $confirmPassword = $_POST["ConfirmPassword"];
    if (
        empty($fullName) || empty($username)
        || empty($email) || empty($password) ||
        empty($confirmPassword)
    ) {
        $alert = "

        Swal.fire({
            icon: 'warning',
            title: 'Missing Information',
            text: 'Please fill in all required fields.', timer: 2000,
            theme: 'dark'

        });

        ";
    }
    // Invalid email

    elseif (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
        $alert = "

        Swal.fire({
            icon: 'warning', title:
```

```
'Invalid Email',
    text: 'Please enter a valid email address.', timer:
    2000,
    theme: 'dark'
});
";
}
// Password mismatch
elseif ($password !== $confirmPassword) {
    $alert = "
    Swal.fire({
        icon: 'error',
        title: 'Passwords Do Not Match',
        text: 'Please make sure both passwords are the same.', timer:
        2000,
        theme: 'dark'
    });
    ";
}
else {
    // Check email
    $sql = "SELECT * FROM users WHERE email = ?";
    $stmt = mysqli_prepare($conn, $sql);
    if (!$stmt) {
        die("Prepare failed: " . mysqli_error($conn));
    }
    mysqli_stmt_bind_param($stmt, "s", $email);
```

```
mysqli_stmt_execute($stmt);

$result = mysqli_stmt_get_result($stmt);
if (mysqli_num_rows($result) > 0) {
    $alert = " Swal.fire({

        icon: 'error',
        title: 'Email Already Registered',

        text: 'An account with this email already exists. Please use another email.', timer: 2000,

        theme: 'dark'
    }); ";

} else {
    // Hash password

    $password = password_hash($password, PASSWORD_DEFAULT);
    // Insert user

    $sql = "INSERT INTO users

        (full_name, username, email, password)

        VALUES (?, ?, ?, ?)";

    $stmt = mysqli_prepare($conn, $sql);
    if (!$stmt) {

        die("Prepare failed: " . mysqli_error($conn));

    }
    mysqli_stmt_bind_param( $stmt, "ssss", $fullName, $username, $email, $password );
    if (mysqli_stmt_execute($stmt)) {
        $alert = "

        Swal.fire({

            icon: 'success',

            title: 'Account Created!',

            text: 'Your TechVolt account has been created successfully.', timer: 2000,

            showConfirmButton: false,
```

```
        theme: 'dark'

    }).then(() => {

        window.location.href = 'index.php';

    });

    ";

} else {
    $alert = "

    Swal.fire({

        icon: 'error',

        title: 'Registration Failed',

        text: 'We could not create your account. Please try again.', theme: 'dark'

    });

    ";

}

}

}

?>
<!DOCTYPE html>

<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Register - TechVolt</title>
    <link rel="stylesheet" href="assets\css\login.css">

    <!-- SweetAlert2 →
    <script src="https://cdn.jsdelivr.net/npm/sweetalert2@11"></script>
</head>
<body>
```

```
<div class="form-container register-container">
  <h2>Create an Account</h2>
  <form action="register.php" method="post">
    <div class="input-box">

      <input id="fullName" name="fullName" type="text" placeholder="">

      <label for="fullName">Full Name</label>

    </div>
    <div class="input-box">

      <input id="username" name="username" type="text" placeholder="">

      <label for="username">Username</label>

    </div>
    <div class="input-box">

      <input id="email" name="email" type="email" placeholder="" >

      <label for="email" class="email">E-mail</label>

    </div>
    <div class="input-box">

      <input id="password" name="password" type="password" placeholder="">

      <label for="password">Password</label>

    </div>
    <div class="input-box">

      <input id="ConfirmPassword" name="ConfirmPassword" type="password" placeholder="" >

      <label for="ConfirmPassword" class="confirm-password">Confirm Password</label>

    </div>
    <button type="submit" class="submit-btn"> Create
      Account</button>
    <div class="extra-link">

      <p>Already have an account?</p>

      <a class="register" href="login.php"> Login
```

➔

```
</div>
</form>
</div>
<section>
  
  <h3>Build Something Amazing. Start Your Journey.</h3>
  <p>Create your TechVolt account and explore the world of
    electronic components, development boards, sensors, modules,
    and more.</p>
</section>
<!-- PHP Alert -->

<?php if (!empty($alert)): ?>
  <script>

    window.addEventListener('DOMContentLoaded', function() {

      <?php echo $alert; ?>

    });

  </script>

<?php endif; ?>
</body>
</html>
```

5.4 SCREENSHOT

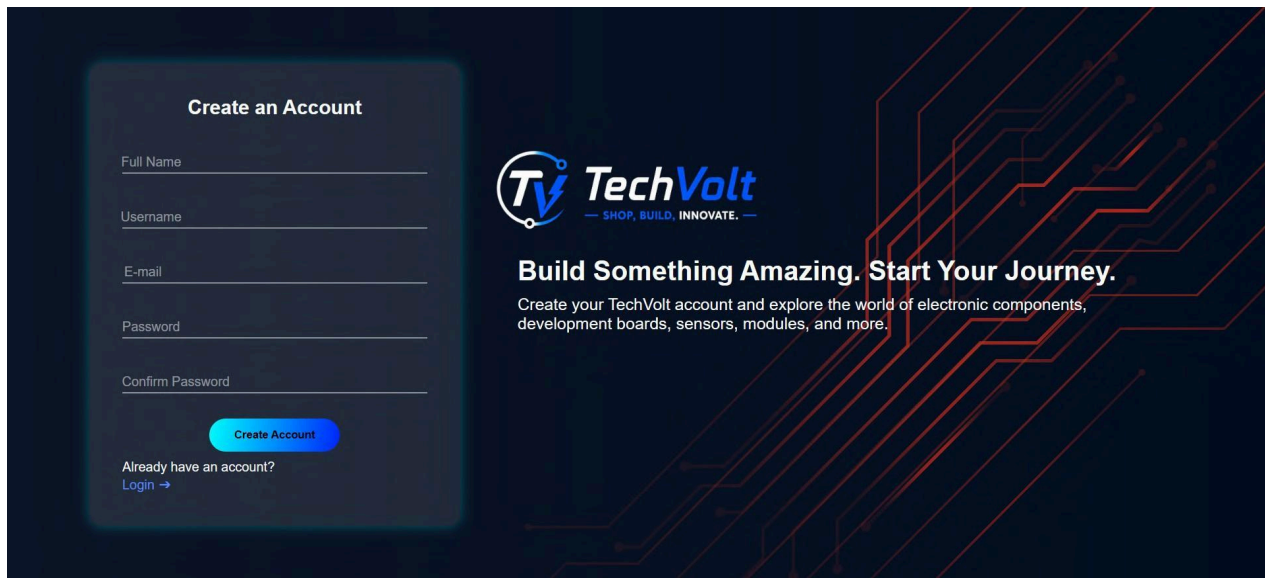
1. REGISTER USER SIDE:

1.1. Login Page



This is the Login page of our website.

1.2 Registration Page



The registration page features a dark blue background with a subtle circuit pattern. On the left, a light blue rounded rectangle contains the registration form. The form has five input fields: 'Full Name', 'Username', 'E-mail', 'Password', and 'Confirm Password'. Below these fields is a blue 'Create Account' button. At the bottom of the form, there is a link 'Already have an account? Login →'. To the right of the form, the TechVolt logo is displayed, consisting of a stylized 'TV' icon and the text 'TechVolt' with the tagline '— SHOP, BUILD, INNOVATE. —' below it. Further right, the heading 'Build Something Amazing. Start Your Journey.' is followed by a paragraph: 'Create your TechVolt account and explore the world of electronic components, development boards, sensors, modules, and more.'

Create an Account

Full Name

Username

E-mail

Password

Confirm Password

Create Account

Already have an account?
[Login →](#)

TechVolt
— SHOP, BUILD, INNOVATE. —

Build Something Amazing. Start Your Journey.

Create your TechVolt account and explore the world of electronic components, development boards, sensors, modules, and more.

This is the Registration page of our website.

1.3. Products Page



Search

0

2

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• Products •

Sort by Categories

Category: Microcontrollers



Microcontrollers

Arduino Uno R3

★★★★★(12)

ATmega328P based development board for electronics projects and prototyping.

Price: ₹799

In Stock

Add to Cart

Wishlist



Microcontrollers

ESP32 DevKit V1

★★★★★(10)

Wi-Fi and Bluetooth enabled development board based on the ESP32 microcontroller.

Price: ₹499

Out of Stock

Add to Cart

Wishlist



Microcontrollers

Arduino Nano

★★★★★(12)

Compact ATmega328P based microcontroller development board.

Price: ₹499

In Stock

Add to Cart

Wishlist



Microcontrollers

ESP8266 NodeMCU

★★★★★(18)

Wi-Fi enabled development board based on ESP8266.

Price: ₹349

In Stock

Add to Cart

Wishlist



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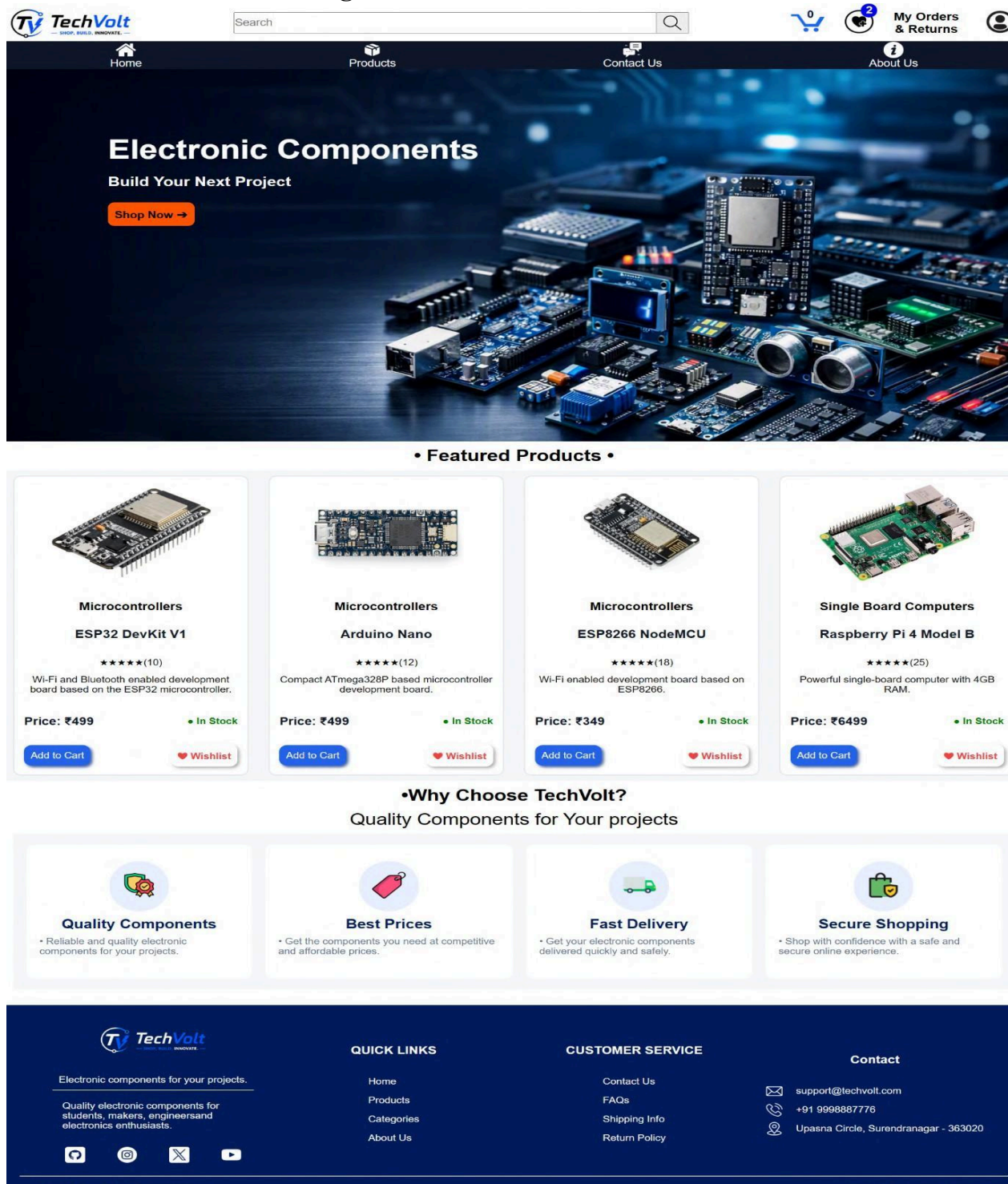
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1.4 Home Page



This is the Products page of our website.

1.5 Product Detail Page







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Arduino Uno R3

★★★★★4.5 (12 Reviews)

₹799

ATmega328P based development board for electronics projects and prototyping.

● In Stock | SKU: TV-MCU-0001

Quantity:

−
1
+

[Add to Cart](#)

[♥ Wishlist](#)

Category: Microcontrollers

Brand: Arduino

Model No: A000066

• Description •

Arduino Uno R3 is a popular ATmega328P-based microcontroller development board designed for electronics projects, prototyping, automation, and learning embedded systems. It provides digital and analog I/O pins for connecting sensors, LEDs, motors, displays, and other electronic components. Its simple programming environment makes it suitable for beginners as well as experienced makers.

• Specification •

Microcontroller	ATmega328P
Operating Voltage	5V
Digital I/O Pins	14
Analog Input Pins	6
Flash Memory	32kb
Clock Speed	16MHz



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This is the Products detail page of our website.

1.6 Cart Page



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Your Cart

3 items in your Cart

All Products



Arduino Uno R3
Microcontrollers
Price: ₹799
✓ In Stock

— 1 +

 Remove



16x2 LCD Display
Displays
Price: ₹149
✓ In Stock

— 2 +

 Remove



LM2596 Buck Converter
Power Modules
Price: ₹69
✓ In Stock

— 1 +

 Remove

Delivery Method

☐ 1 - day Delivery
₹200
• Within 1 working day

☐ 3 - day Delivery
₹100
• Within 3 working days

☒ 7 - day Delivery
₹50
• Within 7 working days

Order Summary

Subtotal:	₹1166
Delivery:	₹50
GST(7%):	₹81.62
Grand Total:	₹1297.62

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1.7 Checkout Page





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Checkout

Complete your order securely

Delivery Address

Saved Addresses

☐ **Bhavik Hakabhal Kher**
9876543210
Limdawali sheri, Sayla, surendranagar, Gujarat - 363430

Full Name:

Mobile Number:

Address:

City: **State:**

Pincode:

Payment Method

☒ **Cash on Delivery**
Pay when your order arrives

☐ **Online Payment**
Pay securely online

Order Summary

	Arduino Uno R3 Price: ₹799 Quantity: 1 Total: ₹799
	16x2 LCD Display Price: ₹149 Quantity: 2 Total: ₹298
	LM2596 Buck Converter Price: ₹69 Quantity: 1 Total: ₹69
Subtotal: ₹1166 Delivery: ₹50 GST(7%): ₹81.62	
Grand Total: ₹1297.62	

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1.7 Order Confirm Page



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Order Placed Successfully!

Your order has been confirmed.
Thank you for shopping with TechVolt.

Order ID: #TV20260024
Estimated Delivery: 30 Sep 2026

Order Detail

	Arduino Uno R3 Quantity: 1	₹799.00
	16x2 LCD Display Quantity: 2	₹298.00
	LM2596 Buck Converter Quantity: 1	₹69.00
Subtotal:		₹1,166.00
Delivery:		₹50.00
GST(7%):		₹81.62
Grand Total:		₹1,297.62

Delivery Information

Name:	Bhavik Hakabhai Kher
Mobile No.	9876543210
Address	Limdawali sheri, Sayla
Payment:	Cash on Delivery
Delivery:	7-Day Delivery
Payment Status:	Pending

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1.7 My Orders & Returns







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My Orders & Returns

Track and manage your orders

Order #TV20260028
Placed on: 23 Sep 2026



PIR Motion Sensor
Qty: 1
₹65

3 more items

Total: ₹7320.36

[View Order Details](#)

Delivered

Order #TV20260027
Placed on: 23 Sep 2026



5V DC Gear Motor
Qty: 1
₹149

3 more items

Total: ₹519.93

[View Order Details](#)

Shipped

Order #TV20260026
Placed on: 23 Sep 2026



ESP8266 NodeMCU
Qty: 1
₹349

3 more items

Total: ₹811.84

[View Order Details](#)

Confirmed

Order #TV20260025
Placed on: 23 Sep 2026



HC-SR04 Ultrasonic Sensor
Qty: 1
₹99

3 more items

Total: ₹990.24

[View Order Details](#)

Cancelled

Order #TV20260023
Placed on: 23 Sep 2026



Arduino Uno R3
Qty: 1
₹799

2 more items

Total: ₹1297.62

[View Order Details](#)

Pending



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1.8 Order Detail Page



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 0
  4
  My Orders & Returns

[← Back to My Orders](#)

Order #TV20260028
 Placed on **23 September 2026**

✓ Delivered

Order Status

✓ Ordered
✓ Processing
✓ Shipped
✓ Delivered

Order Items

Product Image	Product Name & Category	Price	Quantity	Total	Action
	PIR Motion Sensor Sensors	₹65	1	₹65	Return Item
	LDR Light Sensor Module Sensors	₹35	1	₹35	Return Item
	Raspberry Pi 4 Model B Single Board Computers	₹6499	1	₹6499	Return Item
	L298N Dual H-Bridge Motor Driver Module Motors & Actuators	₹149	1	₹149	Return Item

Shipping Address
 Bhavik Hakabhai Kher
 Address:
 Limdawali sheri, Sayla, surendranagar, Gujarat
 9876543210
 Pincode: 363430

Order Summary
 Subtotal: ₹6748.00
 Shipping: ₹100.00
 GST(7%): ₹472.36
 Grand Total: ₹7320.36

Order & Payment Information

Payment Method	Cash on Delivery
Payment Status	Pending
Order Status	Delivered



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This is the order detail page of our website.

1.8.1 Return Modal

Return Product ×

Reason

Select reason ▼

Additional Details

Tell us more about the issue...

Cancel Submit Return

This is the Return Modal in Order detail page of our website.

1.9 Wishlist Page



Search

0

My Orders & Returns

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My WishList

Your Saved Products

4 Items In Your WishList



Microcontrollers

Arduino Uno R3

★★★★ (12)

Price: ₹799 ✓ In Stock

[Add to Cart](#)[Remove](#)



Microcontrollers

ESP32 DevKit V1

★★★★ (10)

Price: ₹499 ✗ Out of Stock

[Add to Cart](#)[Remove](#)



Sensors

HC-SR04 Ultrasonic Sensor

★★★★ (18)

Price: ₹99 ✓ In Stock

[Add to Cart](#)[Remove](#)



Motors & Actuators

MG996R Servo Motor

★★★★ (12)

Price: ₹499 ✗ Out of Stock

[Add to Cart](#)[Remove](#)



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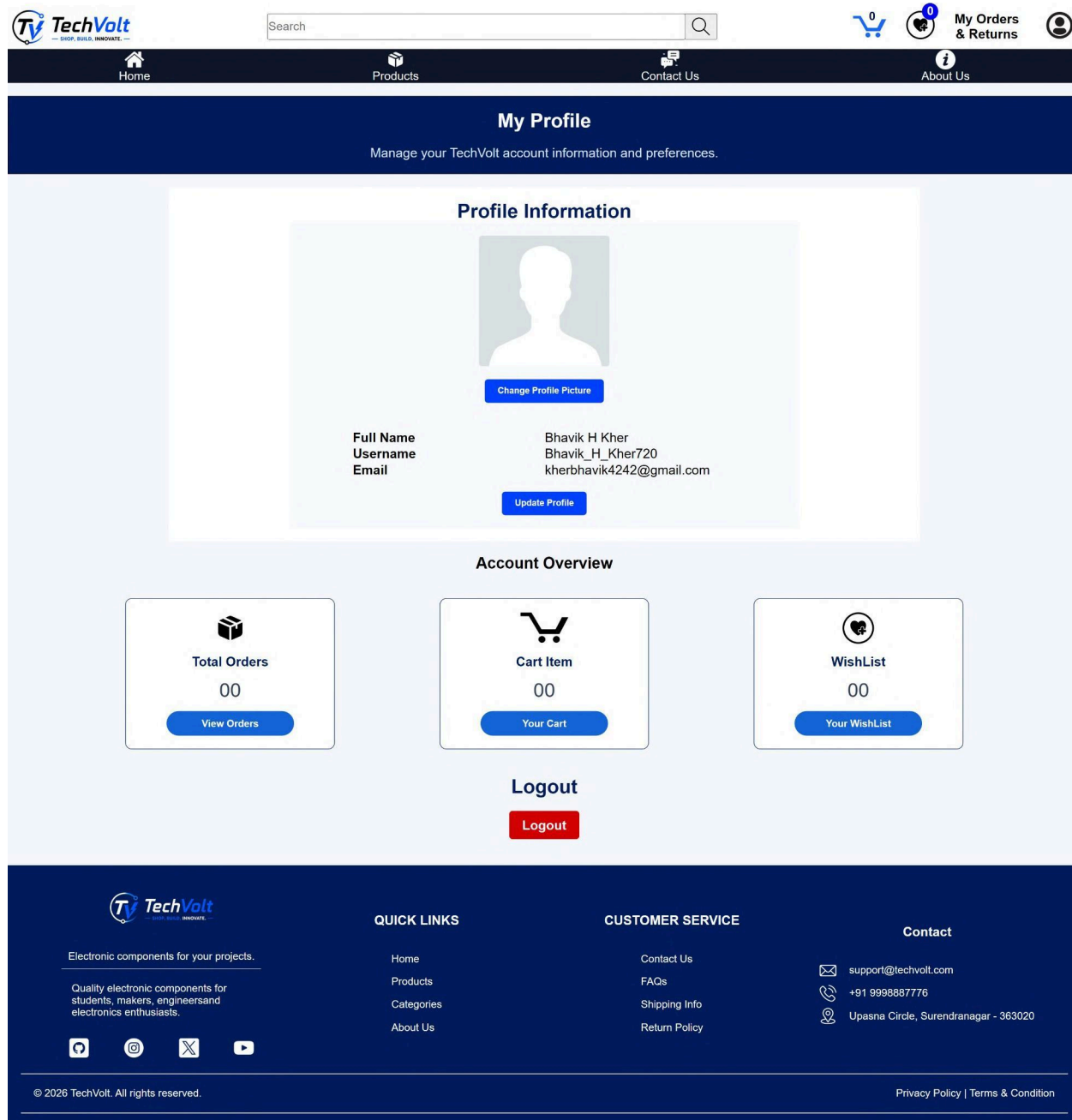
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1.10 Profile Page



This is the Profile Page of our website.

1.11 Search Page





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Search Results

Results for: [Arduino](#)
2 products found



Microcontrollers

Arduino Uno R3

★★★★★ (12)

ATmega328P based development board for electronics projects and prototyping.

Price: ₹799 • In Stock

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Microcontrollers

Arduino Nano

★★★★★ (12)

Compact ATmega328P based microcontroller development board.

Price: ₹499 • In Stock

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[FAQs](#)
[Shipping Info](#)
[Return Policy](#)

Contact

 support@techvolt.com
 +91 9998887776
 Upasna Circle, Surendranagar - 363020

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This is the Search page of our website.

1.12 Contact Us Page





[Home](#)[Products](#)[Contact Us](#)[About Us](#)

CONTACT TECHVOLT

We're here to help.

Have a question? Send us a message and we'll get back to you as soon as possible.

SEND US A MESSAGE

Name:

Email:

Subject:

Message:

[Submit](#)

FREQUENTLY ASKED QUESTIONS

Have questions about TechVolt, products, orders, or shopping? Find helpful answers in our FAQ guide.

[View FAQ Guide →](#)



Electronic components for your projects.

Quality electronic components for students, makers, engineers and electronics enthusiasts.



QUICK LINKS

- [Home](#)
- [Products](#)
- [Categories](#)
- [About Us](#)

CUSTOMER SERVICE

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1.13 About Us Page





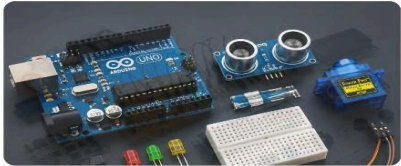


[Home](#)
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About TechVolt

Powering Ideas. Building Projects.

TechVolt is your online destination for quality electronic components, helping students, hobbyists, engineers, and electronics enthusiasts bring their ideas to life.



WHO WE ARE?

Building Ideas with the Right Components

TechVolt is an e-commerce platform for electronic components, created to make it easier to find and purchase the components needed for electronics projects. We offer a wide range of products, from microcontrollers and sensors to displays, ICs, motors, power modules, and other essential components.

[Explore Products →](#)

• WHAT WE OFFER •



Wide Range of Components

Explore components for different electronics projects.



Quality Products

Reliable components for your development and learning.



For Every Project

Find components for learning, prototyping and building.

• OUR MISSION •

"Our mission is to make quality electronic components easy to discover, accessible, and convenient to purchase for students, hobbyists, engineers, and electronics enthusiasts."

Helping you turn ideas into real-world projects with the right components.

BUILT FOR STUDENTS, DESIGNED FOR MAKERS.

TechVolt supports students, hobbyists, engineers, and electronics enthusiasts by providing the right components for learning, prototyping, and innovation.

• READY TO BUILD SOMETHING?

Explore our wide range of electronic components and find everything you need for your next project.

[BROWSE PRODUCTS →](#)



Electronic components for your projects.

Quality electronic components for students, makers, engineers and electronics enthusiasts.



QUICK LINKS

- [Home](#)
- [Products](#)
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- [About Us](#)

CUSTOMER SERVICE

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-  Upasna Circle, Surendranagar - 363020

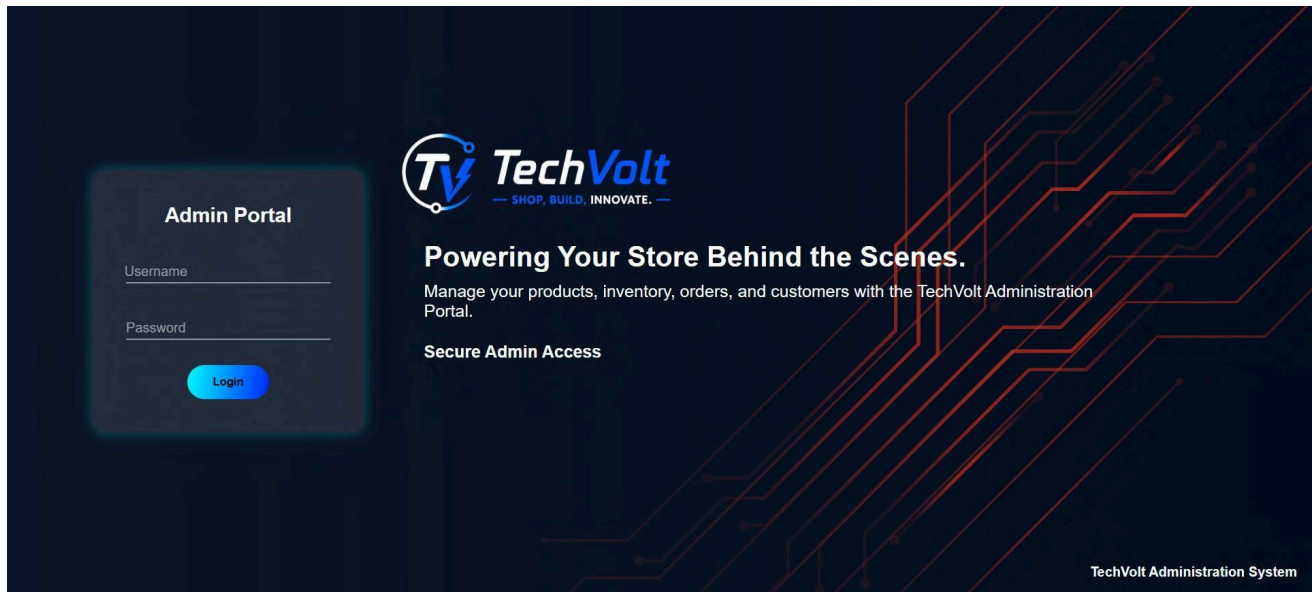
© 2026 TechVolt. All rights reserved.

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This is the About Us page of our website.

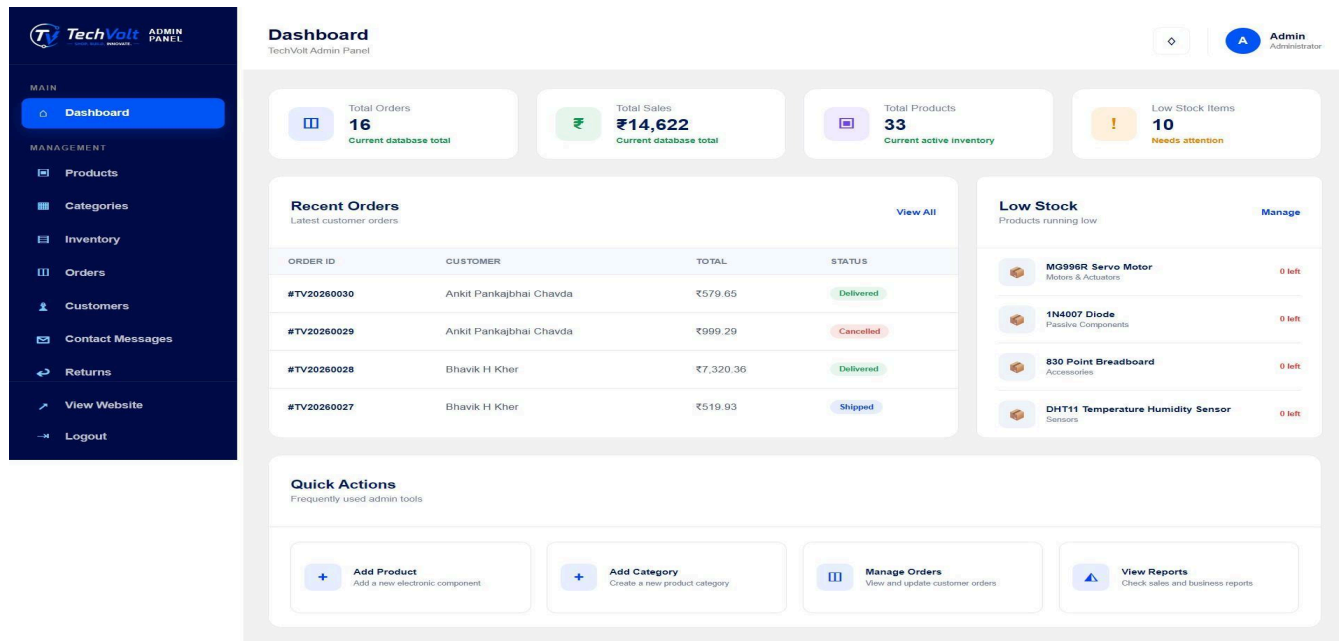
2. ADMIN USER SIDE:

2.1. Admin Login Page



This is the Admin Login Page of the Admin Panel.

2.2. Dashboard Page



This is the Dashboard of the Admin Panel.

2.3. Product-Manage Page

TechVolt ADMIN PANEL

Products
TechVolt Admin Panel

Search products... All Categories Filter

All Products
33 products found Refresh

PRODUCT	CATEGORY	PRICE	STOCK	STATUS	ACTIONS
 Arduino Uno R3 ID: #001	Microcontrollers	₹799	30	In Stock	Edit Delete
 ESP32 DevKit V1 ID: #002	Microcontrollers	₹499	10	In Stock	Edit Delete
 HC-SR04 Ultrasonic Sensor ID: #003	Sensors	₹99	43	In Stock	Edit Delete
 Arduino Nano ID: #004	Microcontrollers	₹499	18	In Stock	Edit Delete
 ESP8266 NodeMCU ID: #005	Microcontrollers	₹349	7	Limited Stock	Edit Delete

Showing 1-5 of 33 products

1 2 3 4 5 6 7

This is the Product Management Page of the Admin Panel.

2.4. Add Product Page

TechVolt ADMIN PANEL

MAIN

- Dashboard

MANAGEMENT

- Products**
- Categories
- Inventory
- Orders
- Customers
- Contact Messages
- Returns
- View Website
- Logout

Products
TechVolt Admin Panel

Admin Administrator

Add Product

Add a new electronic component to your TechVolt store.

[Back to Products](#)

Basic Information

Enter the main details of the product.

Product Name *
Enter product name

Short Description
Enter short product description

Category *
Select Category

Brand
Enter brand name

Model Number
Enter model number

SKU
Enter SKU

Price (₹) *
Enter product price

Stock Quantity *
Enter stock quantity

Product Status *
Active

Product Description
Enter product description

Product Images

Upload product images and select one as the primary image.

[+ Add Images](#)

Add Images
JPG, JPEG or PNG

You can upload multiple images. One image must be selected as the primary product image.

Product Specifications

Add technical specifications of the product.

[+ Add Specification](#)

Specification	Value
Example: Operating Voltage	Example: 5V

[Cancel](#) [Save Product](#)

This is the Add Product Page of the Admin Panel.

2.5. Edit Product Page

The screenshot displays the 'Edit Product' page in the TechVolt Admin Panel. The interface includes a sidebar with navigation options like Dashboard, Products, Categories, Inventory, Orders, Customers, Contact Messages, Returns, View Website, and Logout. The main content area is titled 'Edit Product' and contains three sections: Basic Information, Product Images, and Product Specifications.

Basic Information

Update the main details of the product.

Product Name: Arduino Uno R3

Short Description: ATmega328P based development board for electronics projects and prototyping.

Category: Microcontrollers

Brand: Arduino

Model Number: A000066

SKU: TV-MCU-0001

Price (₹): 799

Stock Quantity: 30

Product Status: Active

Product Description: Arduino Uno R3 is a popular ATmega328P-based microcontroller development board designed for electronics projects, prototyping, automation, and learning embedded systems. It provides digital and analog I/O pins for connecting sensors, LEDs, motors, displays, and other electronic components. Its simple programming environment makes it suitable for beginners as well as experienced makers.

Product Images

Manage existing images or upload new product images.

Four images are shown: 1.jpeg, 1.jpg, 6.jpeg, and 7.jpg. The 7.jpg image is marked as 'Primary'. A dashed box indicates the area to add more images.

Product Specifications

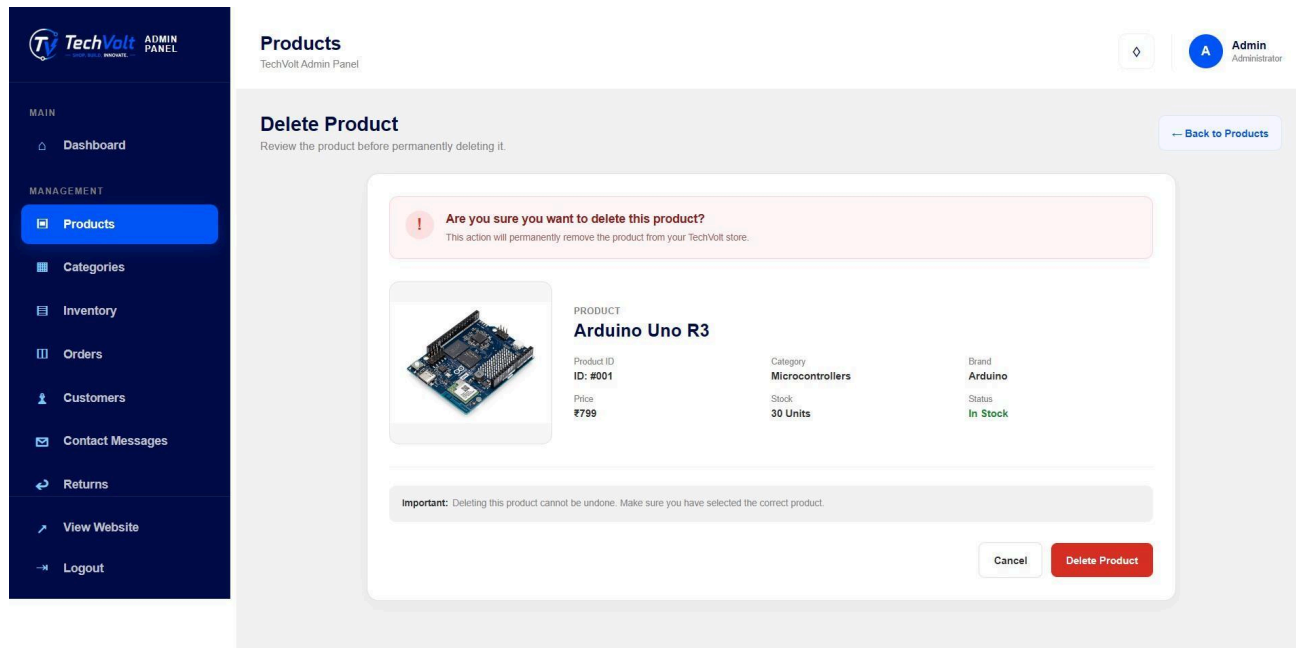
Update the technical specifications.

Specification	Value
Microcontroller	ATmega328P
Operating Voltage	5V
Digital I/O Pins	14
Analog Input Pins	6
Flash Memory	32kb
Clock Speed	16MHz

Buttons: Cancel, Update Product

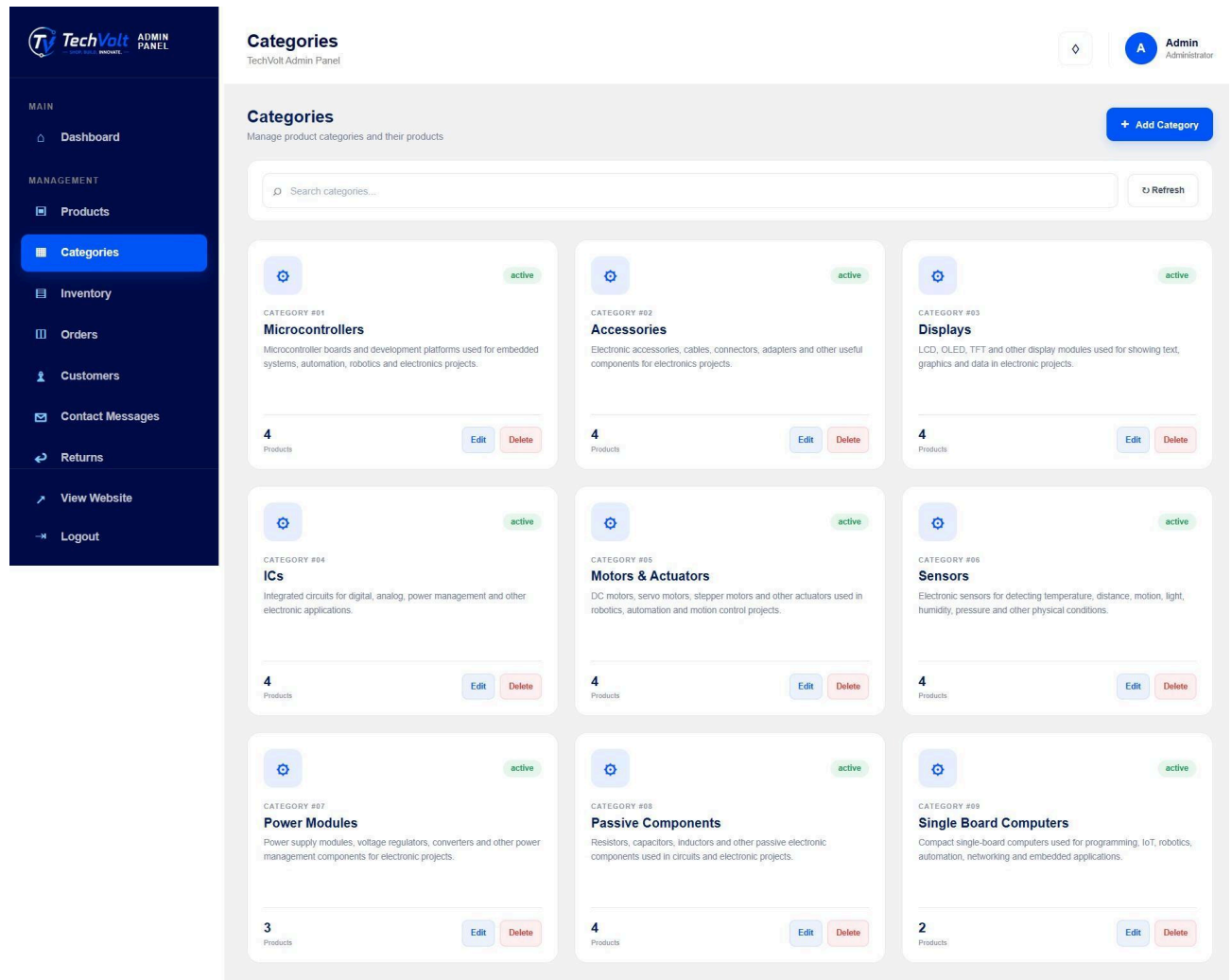
This is the Edit Product Page of the Admin Panel.

2.6. Delete Product Page



This is the Delete Product Page of the Admin Panel.

2.7. Manage Category Page



This is the Manage Category Page of the Admin Panel.

2.8. Add Category Page

TechVolt ADMIN PANEL

MAIN

- Dashboard

MANAGEMENT

- Products
- Categories**
- Inventory
- Orders
- Customers
- Contact Messages
- Returns
- View Website
- Logout

Categories
TechVolt Admin Panel

Add Category
Create a new product category for your TechVolt store.

[Back to Categories](#)

Category Information
Enter the basic details of the new category.

Category Name *
Enter category name

Category Slug
Slug will be generated automatically

Slug is generated automatically from the category name.

Category Description
Enter category description

Category Status
Choose whether this category should be active.

☒ **Active**
Category will be visible in the store.

☐ **Inactive**
Category will remain hidden from the store.

[Cancel](#) [Save Category](#)

This is the Add Category Page of the Admin Panel.

2.9. Edit Category Page

TechVolt ADMIN PANEL

Categories
TechVolt Admin Panel

Edit Category
Update the details of an existing TechVolt category.

[Back to Categories](#)

Category Information
Update the basic details of this category.

Category Name *
Microcontrollers

Category Description
Microcontroller boards and development platforms used for embedded systems, automation, robotics and electronics projects.

Category Status
Update whether this category is active.

☒ **Active**
Category is visible in the store.

☐ **Inactive**
Category will remain hidden from the store.

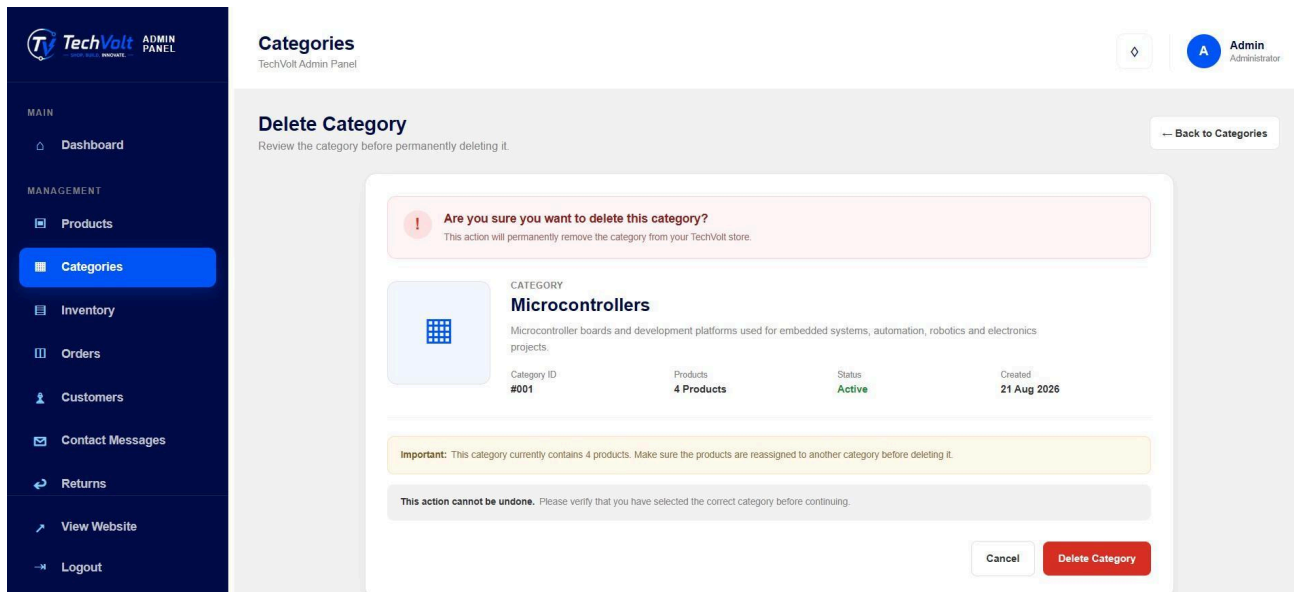
Category Summary
Current information related to this category.

Category ID #001	Products 4 Products	Status Active	Created At 21 Aug 2026
---------------------	------------------------	------------------	---------------------------

[Cancel](#) [Update Category](#)

This is the Edit Category Page of the Admin Panel.

2.10. Delete Category Page



This is the Delete Category Page of the Admin Panel.

2.11. Inventory Page (Stock Management Page)

TechVolt ADMIN PANEL

Inventory
TechVolt Admin Panel

Stock Management
Monitor and manage product stock levels.

Refresh Stock

Summary:

- Total Products: 33
- In Stock: 24
- Low Stock: 4
- Out of Stock: 5

Product Stock
Check current stock levels and update quantities.

Search products... All Categories All Stock Status **Filter**

Product	Category	Price	Current Stock	Status	Last Updated	Action
 Arduino Uno R3 Product ID: 1	Microcontrollers	₹799	30	In Stock	21 Aug 2026	Update
 ESP32 DevKit V1 Product ID: 2	Microcontrollers	₹499	10	In Stock	21 Aug 2026	Update
 HC-SR04 Ultrasonic Sensor Product ID: 3	Sensors	₹99	43	In Stock	21 Aug 2026	Update
 Arduino Nano Product ID: 4	Microcontrollers	₹499	18	In Stock	29 Aug 2026	Update
 ESP8266 NodeMCU Product ID: 5	Microcontrollers	₹349	7	Low Stock	29 Aug 2026	Update

Showing 1 - 5 of 33 products

1 2 3 4 5 6 7

This is the Inventory Page of the Admin Panel.

2.12. Orders Page

TechVolt ADMIN PANEL

Orders
TechVolt Admin Panel

Orders
Monitor customer orders and update their status.

Order Statistics:

- Total Orders: 19
- Pending: 10
- Processing: 0
- Delivered: 4

All Orders
19 orders found

Search order ID or customer... All Status All Payments Filter

Order	Customer	Date	Items	Total	Payment	Status	Action
#TV20260030	AC Ankit Pankajbhai Chavda chavdaank720@gmail.com	24 Sep 2026	1 Item	₹580	COD	Delivered	View
#TV20260029	AC Ankit Pankajbhai Chavda chavdaank720@gmail.com	24 Sep 2026	3 Items	₹999	COD	Cancelled	View
#TV20260028	BK Bhavik H Kher kherbhavik424@gmail.com	23 Sep 2026	4 Items	₹7,320	COD	Delivered	View
#TV20260027	BK Bhavik H Kher kherbhavik424@gmail.com	23 Sep 2026	4 Items	₹520	COD	Shipped	View
#TV20260026	BK Bhavik H Kher kherbhavik424@gmail.com	23 Sep 2026	4 Items	₹812	COD	Confirmed	View

Showing 1 – 5 of 19 orders

This is the Orders Page of the Admin Panel.

2.13. Order Detail Page

The screenshot displays the 'Orders' section of the TechVolt Admin Panel. The left sidebar contains navigation links for Dashboard, Products, Categories, Inventory, Orders (selected), Customers, Contact Messages, Returns, View Website, and Logout. The main content area shows details for 'Order #23', placed on 23 September 2026 at 10:06 PM, with a status of 'Pending'.

Customer Information (Customer details):

- Name: Bhavik H Kher (BH)
- Email: kherbhavik4242@gmail.com
- Phone: 9876543210
- Customer ID: #CUS0010

Shipping Information (Delivery address):

- Name: Bhavik Hakabhai Kher
- Address: Lindawali sheri, Sayla
- Delivery Charge: ₹50.00

Payment Information (Payment details):

- Payment Method: Cash on Delivery
- Payment Status: Pending

Ordered Products (3 products in this order):

Product	Price	Quantity	Total
Arduino Uno R3 Product ID: #1	₹799.00	1	₹799.00
16x2 LCD Display Product ID: #5	₹149.00	2	₹298.00
LM2596 Buck Converter Product ID: #24	₹69.00	1	₹69.00

Order Status (Current order progress):

A progress bar shows four stages: 1. Order Placed (23 September 2026, 10:06 PM) - Completed (green checkmark); 2. Confirmed (Pending); 3. Shipped (Pending); 4. Delivered (Pending).

Order Summary (Payment breakdown):

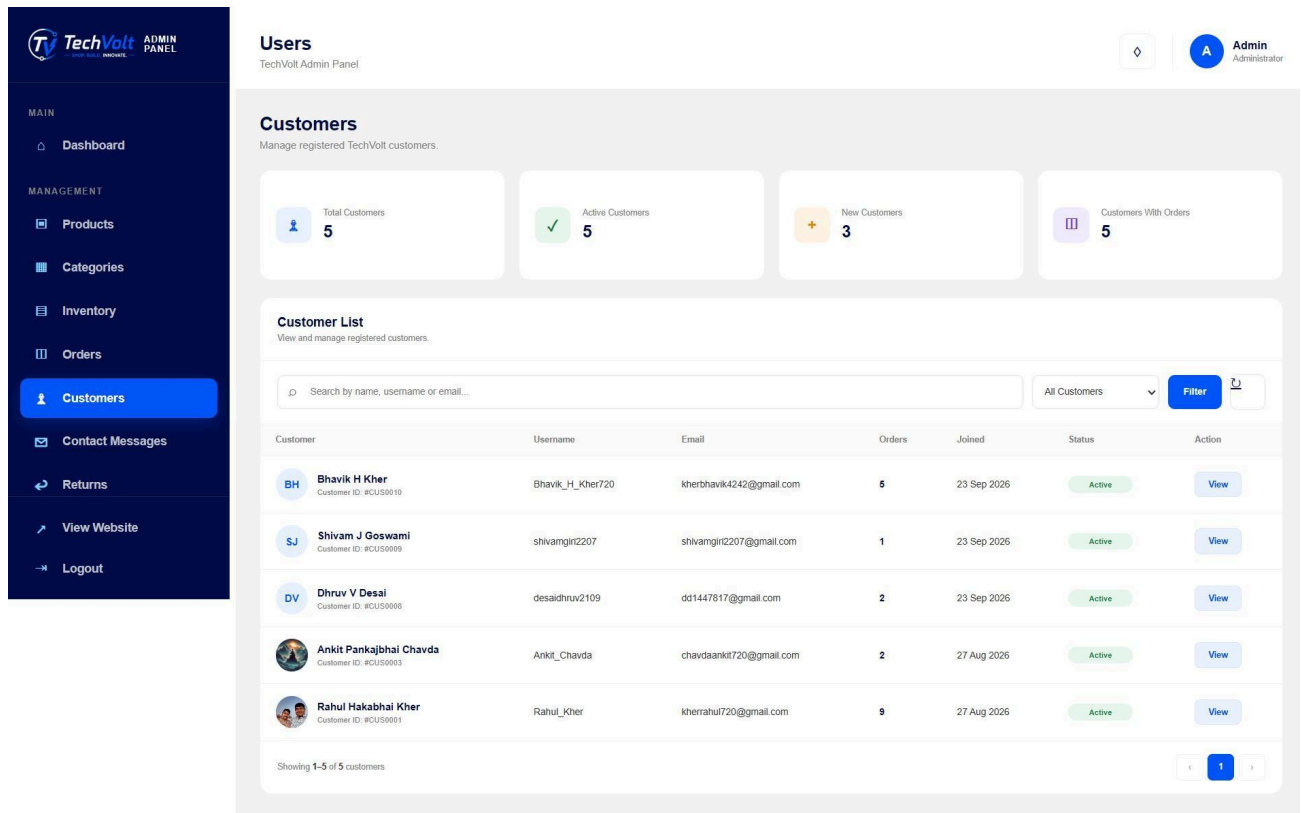
Subtotal	₹1,166.00
Delivery Charge	₹50.00
GST	₹81.62
Total Amount	₹1,297.62

Order Actions (Update the order status from the options below):

Pending Update Status

This is the Order Detail Page of the Admin Panel.

2.14. Customer Page



TechVolt ADMIN PANEL

Users
TechVolt Admin Panel

Customers
Manage registered TechVolt customers.

Customer Statistics:

- Total Customers: 5
- Active Customers: 5
- New Customers: 3
- Customers With Orders: 5

Customer List
View and manage registered customers.

Search by name, username or email...

All Customers Filter

Customer	Username	Email	Orders	Joined	Status	Action
Bhavik H Kher Customer ID: #CUS0010	Bhavik_H_Kher720	kherbhavik4242@gmail.com	5	23 Sep 2026	Active	View
Shivam J Goswami Customer ID: #CUS0009	shivamjir2207	shivamjir2207@gmail.com	1	23 Sep 2026	Active	View
Dhruv V Desai Customer ID: #CUS0008	desaidhruv2109	dd1447817@gmail.com	2	23 Sep 2026	Active	View
Ankit Pankajbhai Chavda Customer ID: #CUS0003	Ankit_Chavda	chavdaankit720@gmail.com	2	27 Aug 2026	Active	View
Rahul Hakabhai Kher Customer ID: #CUS0001	Rahul_Kher	kherrahul720@gmail.com	9	27 Aug 2026	Active	View

Showing 1-5 of 5 customers

This is the Customer Page of the Admin Panel.

2.15. Contact Messages Page

The screenshot displays the TechVolt Admin Panel interface. On the left is a dark blue sidebar with navigation links: Dashboard, Products, Categories, Inventory, Orders, Customers, Contact Messages (highlighted), Returns, View Website, and Logout. The main content area is titled 'Contact Messages' and shows a list of messages received through the Contact Us page. A summary box indicates 'Total Messages: 3'. The messages are listed in a table with columns: CUSTOMER ID, NAME, EMAIL, SUBJECT, MESSAGE, ACTION, and DATE.

CUSTOMER ID	NAME	EMAIL	SUBJECT	MESSAGE	ACTION	DATE
#CU\$0010	Rahul Kher	kherahul720@gmail.com	This is sample long message for this site	Lorem ipsum dolor sit amet consectetur, adipisicing elit...	View Message	24 Sep 2026, 07:44 F
#CU\$0003	Bhavik Hakabhai Kher	kheranu0650@gmail.com	I Love this Website	I love this so much because of this website feature.	View Message	17 Sep 2026, 06:12 F
#CU\$0003	Bhavik Hakabhai Kher	kheranu0650@gmail.com	I Love this Website	I love this so much because of this website feature.	View Message	17 Sep 2026, 06:12 F

This is the Contact Message Page of the Admin Panel.

2.16. Returns Page

TechVolt ADMIN PANEL

Returns
TechVolt Admin Panel

Returns
Manage customer return requests and their status.

Summary:

- Total Returns: 4
- Pending: 1
- Approved: 1
- Rejected: 1

Return Requests
Review customer return requests.

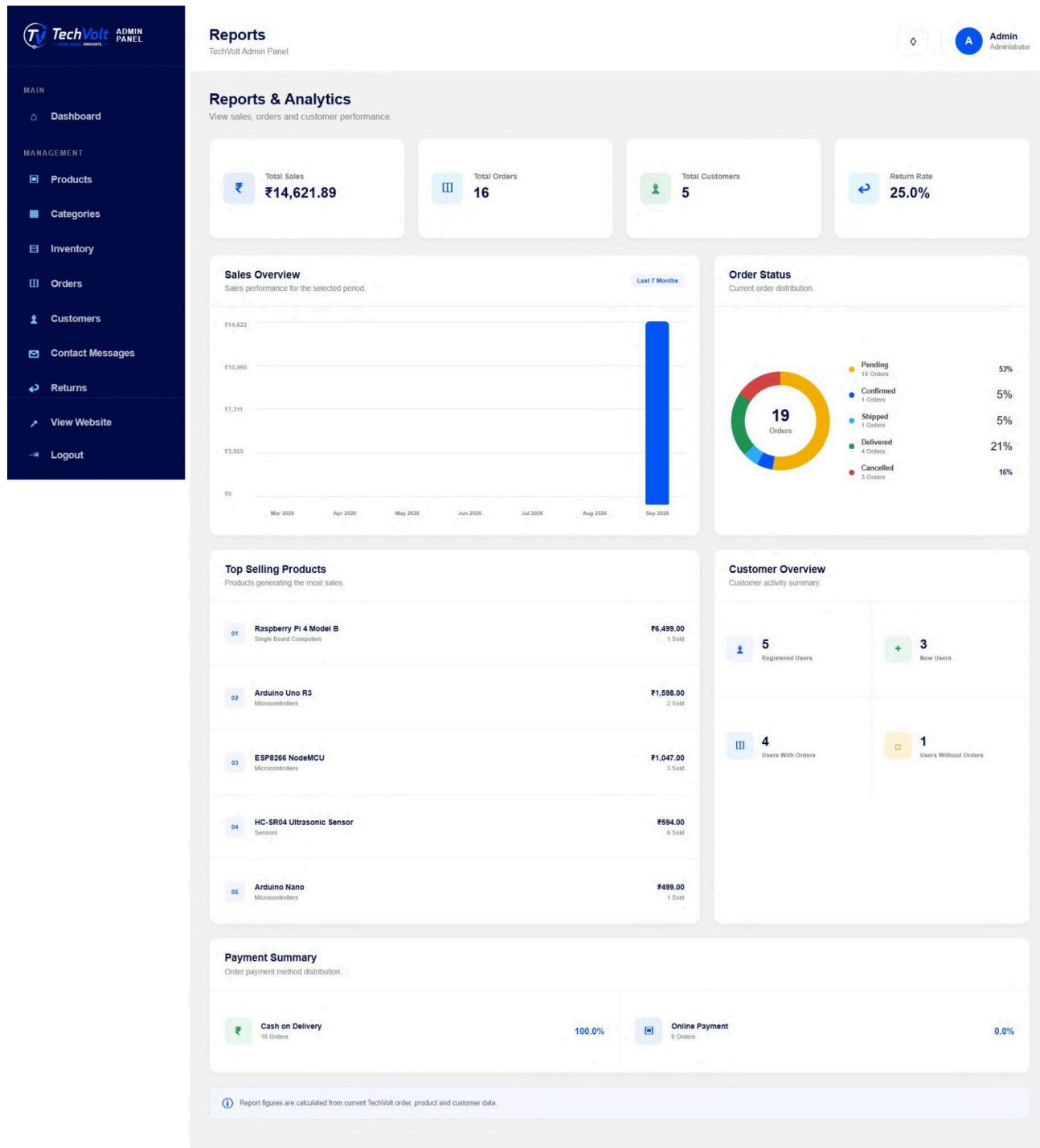
Search by order ID, customer or product... All Status Filter

RETURN ID	ORDER ID	CUSTOMER	PRODUCT	REASON	REQUEST DATE	STATUS	ACTION
#RET004	#TV000012	DV Dhruv V Desai dd1447817@gmail.com	16x2 LCD Display	damaged	23 Sep 2026	Requested	View
#RET003	#TV000012	DV Dhruv V Desai dd1447817@gmail.com	0.96 inch OLED Display	damaged	23 Sep 2026	Approved	View
#RET002	#TV000012	DV Dhruv V Desai dd1447817@gmail.com	Arduino Nano	not-as-described	23 Sep 2026	Rejected	View
#RET001	#TV000010	Rahul Hakabhai Kher kherahul720@gmail.com	20x4 LCD Display	damaged	03 Sep 2026	Completed	View

Previous 1 Next

This is the Returns Page of the Admin Panel.

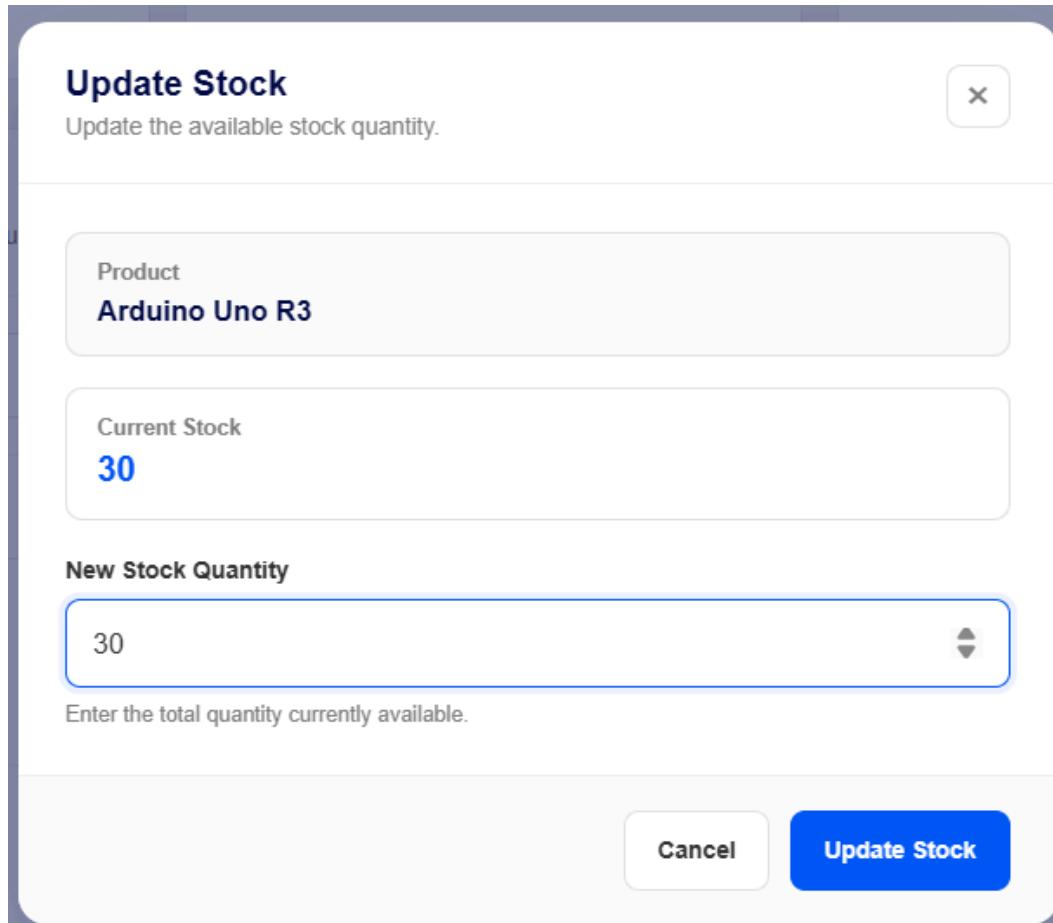
2.17. Reports Page



This is the Reports Page of the Admin Panel.

2.18. Modals

2.18.1 Update Stock



The image shows a modal window titled "Update Stock" with a close button (X) in the top right corner. Below the title is a subtitle "Update the available stock quantity." The modal contains three main sections: 1. "Product" section with a label "Product" and a value "Arduino Uno R3". 2. "Current Stock" section with a label "Current Stock" and a value "30". 3. "New Stock Quantity" section with a label "New Stock Quantity" and a text input field containing "30". Below the input field is a hint text "Enter the total quantity currently available." At the bottom right of the modal are two buttons: "Cancel" and "Update Stock".

Update Stock ×

Update the available stock quantity.

Product
Arduino Uno R3

Current Stock
30

New Stock Quantity

30 ⬆️⬆️

Enter the total quantity currently available.

Cancel **Update Stock**

This is the Update Stock Modal for update stock of specific product in the Admin Panel.

2.18.2 Customer Detail

Customer Details

View registered customer information.

BH

Bhavik H Kher
@Bhavik_H_Kher720

Active

Account Information

Customer ID #CUS0010	Joined 23 Sep 2026
Total Orders 5	Total Spent ₹9,949.75

Contact Information

@

Email
kherbhavik4242@gmail.com

Close

This is the Customer Detail Modal for viewing a customer detail in the Admin Panel.

2.18.3 Contact Messages

Contact Message

NAME

Rahul Kher

EMAIL

kherrahul720@gmail.com

SUBJECT

This is sample long message for this site

DATE

24 Sep 2026, 07:44 PM

MESSAGE

Lorem ipsum dolor sit amet consectetur, adipisicing elit. Nostrum, explicabo cumque nisi non earum aperiam incidunt rem deleniti culpa, eveniet dolorum eum qui repudiandae fugit quas natus quasi ipsa nam! Consequuntur quibusdam expedita harum quisquam aspernatur debitis consectetur, atque architecto repellendus soluta ut, maiores modi laudantium quis! Distinctio voluptatem, qui eum dolorum sed nobis quaerat culpa, minima possimus pariatur excepturi. Porro eligendi itaque repellat dolores quasi debitis rerum rem libero nam placeat eum autem exercitationem, mollitia nisi optio beatae commodi quas qui architecto blanditiis quam similique accusamus. Aliquid, temporibus unde! Fuga quasi delectus eligendi commodi aliquam dolor velit reiciendis atque! Explicabo veniam fuga cumque nam voluptas, debitis nobis numquam? Sint eaque velit harum distinctio? Officiis saepe voluptates voluptatum perspiciatis labore. In amet earum illum, deserunt aperiam placeat modi, recusandae eveniet, fugit accusamus ipsa nihil. Ut, vero dolorum quidem illo culpa expedita dolor corrupti modi quod? Qui consequuntur assumenda sunt ratione.

Close

This is the Contact Message Modal for viewing customer messages which sent by customer in the Admin Panel.

2.18.4 Returns - Modal

Return Details

View complete return request information.

Return Information

Return ID #RET004	Return Date 23 Sep 2026
Order ID #TV000012	Status Requested

Customer Information

DV

Dhruv V Desai
dd1447817@gmail.com

Product Information

16x2 LCD Display Quantity: 1	₹149
--	-------------

Close

Reject Return

Approve Return

This is the Return Modal for approving or rejecting a return requests in the Admin Panel.

TESTING

Website testing is a basic component of website quality assurance and addresses a systematic review of the requirements, design and coding of the TechVolt website. Testing is an important stage of the software development process which includes testing the system using different test data. Preparation of test data plays an important role in system testing. After preparing the test data, the system under review is tested using those test data. If testing is conducted properly, it helps to identify the errors and defects present in the website. Testing also demonstrates that the functions of the TechVolt website are working according to the specified requirements and that the required performance and functionality have been achieved. The information collected during testing provides a good indication of the reliability of the TechVolt website and gives an overall indication of the quality of the system.

The development of the TechVolt website involves a series of development activities where opportunities for human errors are possible. Errors may start to occur at the initial stage of the development process where the requirements may be incorrectly or incompletely identified, as well as during the later design, coding and implementation stages. Due to human limitations in performing and communicating tasks perfectly, website development is followed by a quality assurance activity. Therefore, testing is performed on different modules of the TechVolt website to verify that the system works correctly and satisfies the specified requirements. The testing process includes checking the Registered User module and Admin module, database operations, product management, cart, wishlist, checkout, orders, inventory, returns and other important functionalities of the website.

6.1 Test Cases

In website engineering, the most common definition of a test case is a set of conditions or variables under which a tester determines whether a requirement or use case of an application is partially or fully satisfied. It may take several test cases to determine

that a particular requirement is completely satisfied. In order to fully test whether all the requirements of the TechVolt website are met, different test cases are prepared and executed for various modules of the system. The test cases include valid as well as invalid inputs to verify the expected behaviour of the website. The actual results obtained during testing are compared with the expected results, and the test case is marked as Pass when the system produces the expected output.

Login Test

Sr. No.	Test Case	Expected Results	Actual Result
1.	Try to login with empty username and password fields	An error message is displayed asking the user to enter the required fields.	Pass
2.	Enter an invalid username or email and password	An error message is displayed: "Invalid username or password".	Pass
3.	Enter valid username and password	User is successfully logged in and redirected to the website.	Pass
4.	Login using registered email and correct password	User is successfully logged in.	Pass
5.	Try to login using an inactive account	User is not allowed to login.	Pass

Table 6.2: Login Test Cases

Register Test

Sr. No.	Test Case	Expected Results	Actual Result
1.	Try to register with empty required fields	An error message is displayed asking the user to enter the required information.	Pass
2.	Enter an invalid email address	An error message is displayed for invalid email format.	Pass
3.	Enter a password and different confirm password	An error message is displayed indicating that passwords do not match.	Pass
4.	Enter an already registered username or email	An error message is displayed indicating that the account already exists.	Pass
5.	Enter valid registration details	User account is created successfully.	Pass

Table 6.3: Register Test Cases

Product Test

Sr. No.	Test Case	Expected Results	Actual Result
1.	Open the Products page	Available active products are displayed.	Pass
2.	Search for an existing product	Matching product is displayed.	Pass
3.	Search for a non-existing product	No matching product message is displayed.	Pass
4.	Open product details	Product name, image, price, stock and specifications are displayed.	Pass
5.	Open an inactive product through direct URL	Product is not displayed to the customer.	Pass

Table 6.4: product Test Cases

LIMITATIONS & FUTURE EXTENSIONS

7.1 LIMITATIONS:

- Require more accuracy in maintaining product, stock and order information.
- The website provides limited functionality as it is developed as an academic project.
- In this project, no OTP system is available for user registration or verification.
- In this project, no notification, message and E-mail facility is available.
- In this project, advanced online payment and payment gateway facilities are limited.
- In this project, product review and rating functionality is not available.

7.2 FUTURE EXTENSIONS:

In an advanced technology environment, we can provide more facilities and better services to the users.

- We can provide online payment gateway integration for faster and more convenient order processing.
- In the future, we can add an OTP system for user registration and account verification.
- In future, notification, message and E-mail facilities can be added for order updates, offers and other important information.
- We can add product reviews and ratings so that users can share their experience about purchased products.
- In the future, advanced product recommendation functionality can be added according to user interests and previous orders.
- We can provide a mobile application for TechVolt to make electronic component shopping more convenient for users.
- More advanced reports and analytics can be added to help the administrator understand sales, products and customer activities.
- In future, additional security features can be implemented to provide better protection for user and order information.

CONCLUSION AND REFERENCES

8.1 CONCLUSION

It has been a great pleasure for us to work on this exciting and challenging project. This project proved useful for us as it provided practical knowledge about web application development using HTML, CSS, JavaScript, PHP and MySQL, along with database management and server-side programming.

The project “**TechVolt – Electronic Components E-Commerce Website**” helped us understand the complete development process of an e-commerce website. During the development of this project, we gained practical knowledge about user registration and login, product browsing and searching, product details, cart and wishlist management, checkout and order processing. We also learned about developing an Admin Panel for managing products, categories, inventory, customers, orders, returns and reports.

It also provided knowledge about database connectivity, data management and the technologies used for developing web-enabled applications. The project helped us understand how frontend and backend technologies work together with a MySQL database to provide a complete web application. This practical experience will provide better opportunities and guidance for us in future web development projects and will help us in developing projects independently.

8.2 REFERENCES

Websites:

- [PHP Official Documentation](#)
- [MySQL Documentation](#)
- [MDN Web Docs](#)
- [JavaScript Documentation](#)
- [SweetAlert2 Documentation](#)
- [Draw.io / diagrams.net](#)